
From Noise to Diversity: Random Embedding Injection in LLM Reasoning

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Abstract

Recent *soft prompt* methods, which prepend or insert trainable continuous embeddings into LLM inputs as virtual tokens, optimize these embeddings to enhance mathematical reasoning. However, the mechanism by which embedding injection itself affects model behavior, independent of optimization, remains unexplored. We investigate Random Soft Prompts (RSPs), continuous vectors sampled from the model’s token embedding distribution and injected without any training. Empirically, RSPs increase early-stage token entropy while generation stabilizes thereafter, and combining RSPs with temperature sampling yields higher trajectory diversity as measured by Pass@ n . Theoretically, we provide two complementary analyses: an energy-based framework showing that RSP introduces Langevin-like perturbations on the Hopfield energy landscape with natural dual decay across layers and generation steps, and a distributional analysis showing that RSP induces stochastic token rankings and expands the output support beyond temperature sampling. We further discuss the implications for GRPO training, where trajectory diversity is critical for learning signal quality.

1 Introduction

As large language models (LLMs) scale to billions of parameters, full fine-tuning becomes increasingly expensive. This has motivated parameter-efficient methods that adapt frozen models by introducing a small number of trainable parameters [Hu et al., 2022, Housby et al., 2019]. Among these, *soft prompts* prepend or insert learnable continuous embeddings into the input sequence, steering model behavior through the attention mechanism [Li and Liang, 2021, Lester et al., 2021]. This paradigm has been actively adopted for enhancing mathematical reasoning in LLMs [Kang et al., 2026, Xu et al., 2025, Ye et al., 2026, Hao et al., 2025, Zhang et al., 2026]. Despite their diverse designs, these methods share a common structure: they inject continuous embeddings that lie outside the pretrained vocabulary and optimize them to improve downstream performance.

However, we find that optimization is not a necessary condition for these effects. We observe that injecting randomly initialized, untrained continuous embeddings can also shift the output distribution. For instance, applying a chat template to a base model not fine-tuned with such format causes a prompt-format mismatch that degrades reasoning capability, but appending random embeddings to such mismatched inputs mitigates this degradation. This suggests that injecting out-of-vocabulary continuous embeddings itself, independent of any learned content, plays a non-trivial role in shaping model behavior. Existing studies primarily evaluate *soft prompt* methods through end-task accuracy, and the mechanisms by which embedding injection influences the output distribution remain insufficiently explored.

In this work, we investigate the role of Random Soft Prompts (RSPs), continuous vectors sampled from the Gaussian distribution of the model’s token embedding matrix and injected without any

training. Empirically, RSPs increase early-stage token entropy while the output stabilizes in later generation stages, and combining RSPs with temperature sampling yields more diverse reasoning trajectories as measured by Pass@ n scaling. Theoretically, we analyze RSPs from two perspectives: an energy-based framework showing that RSP introduces controlled stochastic perturbations with natural decay through the attention mechanism, and a distributional analysis showing that RSP induces stochastic token rankings and expands the output support beyond what temperature sampling can achieve. We further discuss the implications of these findings for Group Relative Policy Optimization (GRPO) [Shao et al., 2024], where trajectory diversity is essential for effective learning signals.

Our contributions are as follows:

- We empirically analyze how RSPs affect LLM generation, showing that RSPs increase early token entropy and produce more diverse reasoning trajectories when combined with temperature sampling.
- We provide theoretical analyses from two perspectives: an energy-based framework explaining why RSP can perturb the output distribution without degrading performance, and a distributional analysis showing that RSP induces stochastic token rankings and expands the output support beyond the reach of temperature sampling.
- We discuss the implications of RSP-induced diversity for GRPO training and argue that RSP provides exploration gains complementary to existing temperature-based approaches.

2 Background

2.1 Soft Prompt Methods for LLM Reasoning

Soft prompts emerged as a parameter-efficient alternative to full fine-tuning. Prefix-Tuning [Li and Liang, 2021] introduced the paradigm by prepending trainable continuous vectors to every transformer layer, Prompt Tuning [Lester et al., 2021] simplified this to input-layer-only tuning and showed that it matches full fine-tuning at sufficient scale, and P-tuning [Liu et al., 2024] extended the approach with an LSTM-based prompt encoder to model inter-position dependencies.

More recently, *soft prompts* have been actively applied to LLM reasoning. TTSV [Kang et al., 2026] optimizes prefix embeddings at test time via entropy minimization, steering the model toward high-certainty states to activate latent reasoning capabilities. SoftCoT [Xu et al., 2025] replaces explicit chain-of-thought tokens with soft tokens generated by an assistant model, leveraging the stronger representational capacity of continuous representations over discrete language. LTPO [Ye et al., 2026] optimizes latent thought embeddings per test instance through policy gradient with a confidence-based intrinsic reward. COCONUT [Hao et al., 2025] feeds the model’s own hidden states back as input embeddings, enabling reasoning in a continuous latent space. MemGen [Zhang et al., 2026] demonstrates that embedding vectors can serve as experience memory by constructing latent token sequences that enrich the model’s reasoning. Pause tokens [Goyal et al., 2024] insert learnable tokens into the input to provide additional computation steps.

These approaches all inject out-of-vocabulary continuous embeddings optimized for task-specific objectives. Because evaluation centers on end-task accuracy, the contribution of the injection itself and that of the optimization are not separated, and the mechanisms by which such out-of-vocabulary continuous embeddings shape model behavior remain a separate and underexplored question.

2.2 Noise Injection in Neural Networks

Neural networks have incorporated noise at various stages. During training, dropout [Srivastava et al., 2014] randomly deactivates hidden units to prevent overfitting, and NEFTune [Jain et al., 2024] adds noise to token embeddings during instruction fine-tuning, reporting improvements in instruction following. These methods use noise as a regularizer and do not apply perturbations at inference time.

At inference time, perturbation-based approaches primarily target robustness and safety. Randomized smoothing [Cohen et al., 2019] injects Gaussian noise into inputs to obtain certified robustness guarantees. In the LLM setting, SmoothLLM [Robey et al., 2025] perturbs input prompts at the character level to defend against adversarial jailbreaking. These methods require multiple noisy forward passes with output aggregation, and target prediction stability rather than output diversity.

Table 1: Accuracy (%) across model configurations and benchmarks. RSP reports the best result across injection positions (Prefix, Infix, Suffix). Full per-position breakdown is in Appendix A. Values in parentheses indicate the difference from the baseline.

Model	MATH-500		GSM8K		AIME24	
	Baseline	RSP	Baseline	RSP	Baseline	RSP
<i>Instruct Models</i>						
LLaMA-3.1-8B-Inst	52.20	49.40 (−2.8)	85.75	86.73 (+1.0)	6.67	10.00 (+3.3)
Qwen2.5-Math-7B-Inst	83.20	84.20 (+1.0)	95.45	95.68 (+0.2)	13.33	16.67 (+3.3)
Qwen2.5-Math-1.5B-Inst	73.00	75.20 (+2.2)	85.29	85.75 (+0.5)	10.00	20.00 (+10.0)
<i>Base Models (Raw Text)</i>						
Qwen2.5-Math-7B	72.20	71.60 (−0.6)	84.15	84.31 (+0.2)	6.67	16.67 (+10.0)
Qwen2.5-Math-1.5B	61.40	64.40 (+3.0)	77.86	74.30 (−3.6)	16.67	10.00 (−6.7)
<i>Base Models + ChatML (Format Mismatch)</i>						
Qwen2.5-Math-7B	52.20	70.40 (+18.2)	58.30	75.97 (+17.7)	23.33	23.33 (+0.0)
Qwen2.5-Math-1.5B	34.40	63.40 (+29.0)	37.45	67.02 (+29.6)	13.33	20.00 (+6.7)

In reinforcement learning, NoisyNet [Fortunato et al., 2018] injects learnable noise into network weights to improve exploration, demonstrating that the location and structure of noise injection affect exploration quality. This motivation is closest to RSPs, which similarly target diversity, but differ in injection target (input embeddings vs. network weights) and require no learned noise parameters.

RSPs differ from these approaches in the following respects. **(1)** RSPs operate solely at inference without any training, whereas dropout and NEFTune inject noise during training, modifying the resulting model parameters accordingly. **(2)** RSPs preserve the original input and append fresh embedding vectors in a single forward pass, whereas robustness methods corrupt the input and aggregate over multiple noisy passes. **(3)** RSPs use purely random vectors sampled from the pretrained embedding statistics, whereas NoisyNet learns noise parameters through optimization.

3 Experimental Analysis

3.1 Random Soft Prompts: Definition and Experimental Setup

Let $\mathbf{W}_E \in \mathbb{R}^{V \times d}$ denote the token embedding matrix of a pretrained LLM, where V is the vocabulary size and d is the hidden dimension. We define an RSP $\mathbf{H} = [h_1, \dots, h_L] \in \mathbb{R}^{L \times d}$ as a sequence of L continuous vectors, where each vector is independently sampled from an isotropic Gaussian distribution whose mean and variance are matched to those of the pretrained token embedding matrix:

$$h_j \sim \mathcal{N}(\mu_E, \sigma_E^2 \mathbf{I}), \quad j = 1, \dots, L, \quad (1)$$

with $\mu_E = \frac{1}{Vd} \sum_{i,k} [\mathbf{W}_E]_{i,k}$ and $\sigma_E = \text{std}(\mathbf{W}_E)$. Let $\mathbf{X} = [x_1, \dots, x_T] \in \mathbb{R}^{T \times d}$ denote the embedding sequence of the original input. The RSP-augmented input $\tilde{\mathbf{X}}$ is constructed by concatenating $\mathbf{H}$ with $\mathbf{X}$ at one of three positions without modifying the original prompt text: *prefix* ($[\mathbf{H}; \mathbf{X}]$), *infix* ($\mathbf{H}$ inserted within $\mathbf{X}$), or *suffix* ($[\mathbf{X}; \mathbf{H}]$), following the injection strategies adopted in prior work [Kang et al., 2026, Xu et al., 2025, Ye et al., 2026, Zhang et al., 2026]. Sampling a new $\mathbf{H}$ for each batch ensures that the results are not dependent on any realization of the random embeddings.

We evaluate on mathematical reasoning benchmarks: MATH-500 [Lightman et al., 2024], GSM8K [Cobbe et al., 2021], and AIME24, using LLaMA-3.1-8B-Instruct [Grattafiori et al., 2024] and the Qwen2.5-Math model family [Yang et al., 2024] across instruct models, base models, and base models with mismatched chat templates. All experiments use greedy decoding to isolate the effect of RSPs from sampling stochasticity. Full experimental details are provided in Appendix A.

3.2 How Does Untrained RSP Compare to Baselines & Optimized Soft Prompts?

We first examine how injecting random embeddings affects the model’s reasoning performance. Table 1 reports the accuracy for each model configuration, using the best RSP result across injection positions (Prefix, Infix, Suffix) and $L \in \{10, 15, 20\}$ for each benchmark.

Table 2: Comparison with existing soft prompt methods (TTSV, SoftCoT, LTPO) under unified evaluation. Baseline and RSP values are from Table 1. **Bold** denotes the best result and underline denotes the second best for each model–benchmark pair.

Model	Benchmark	Baseline	RSP	TTSV	SoftCoT	LTPO
Qwen2.5-Math-7B-Instruct	MATH-500	83.20	84.20	<u>83.80</u>	82.80	83.00
	GSM8K	<u>95.45</u>	95.68	<u>95.30</u>	95.00	94.84
	AIME24	13.33	<u>16.67</u>	23.30	<u>16.67</u>	13.33
Qwen2.5-Math-1.5B-Instruct	MATH-500	73.00	<u>75.20</u>	<u>75.20</u>	76.00	72.40
	GSM8K	85.29	85.75	<u>85.70</u>	84.46	84.53
	AIME24	<u>10.00</u>	20.00	6.70	6.67	<u>10.00</u>

For instruct models and base models with their native prompt formats, RSP injection maintains or slightly shifts baseline performance within a few percentage points. Most strikingly, under prompt-format mismatch — where base models receive ChatML templates they were not trained on — RSP yields pronounced improvements of up to +29 pp. These gains are difficult to attribute to simple perturbation effects, as noise that merely raises model uncertainty would further degrade performance.

We compare RSPs with prior soft prompt methods (TTSV, SoftCoT, LTPO) that optimize their embeddings for distinct objectives, under a unified evaluation protocol reported in Table 2. All methods are re-evaluated with the same unified extraction pipeline (Appendix A.2), enabling a format-agnostic comparison despite each method’s distinct prompt format and grading scheme. The results place RSPs on par with the optimized methods without requiring any training.

3.3 How Does RSP Affect the Output Distribution?

We analyze how RSP injection changes the model’s output distribution on Qwen2.5-Math-1.5B-Instruct with MATH-500, measuring per-token entropy, top-1 probability, and varentropy across the generation process. Figure 1 compares these metrics for the first 5% of generation steps across RSP variants and prior soft prompt methods (LTPO, SoftCoT, TTSV). Full statistics are in Appendix A.4.

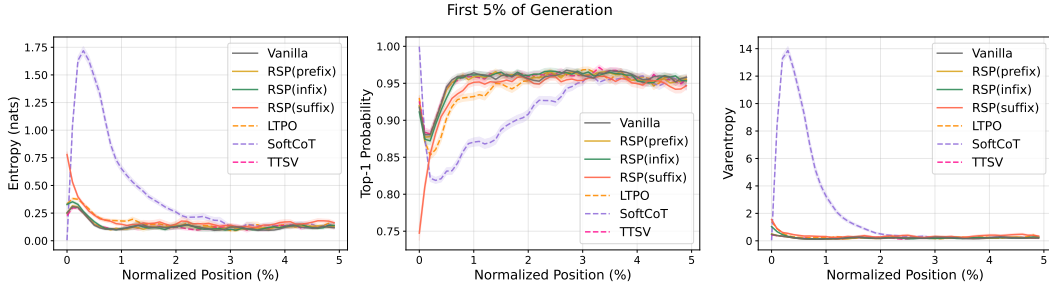


Figure 1: Mean entropy, top-1 probability, and varentropy during the first 5% of generation steps (Qwen2.5-Math-1.5B-Instruct, MATH-500). Shading indicates ± 1 SEM. Solid lines: RSP variants and the baseline. Dashed lines: prior soft prompt methods (LTPO, SoftCoT, TTSV).

These results show that RSP modifies the early-stage output distribution across all injection positions: entropy increases (the token distribution flattens), top-1 probability decreases (commitment to a dominant token weakens), and varentropy increases (indicating multimodal distributions [Ahmed et al., 2026]). High-entropy positions are known to act as critical forking points that determine reasoning trajectories [Wang et al., 2025], so this shift moves the model toward a state where multiple reasoning paths can coexist. The change is localized to the initial reasoning stage. In the middle and final portions of generation, all three metrics converge to baseline levels.

Although existing soft prompt methods are optimized for different objectives, they share a common pattern of increasing early-stage entropy. Notably, TTSV, which optimizes soft prompts using a reward that reduces the mean entropy of reasoning trajectories, still exhibits early entropy comparable to the baseline. This suggests that elevating early-stage entropy is an inherent effect of injecting non-pretrained continuous embeddings, independent of the optimization objective.

3.4 Does RSP Induce Trajectory Diversity?

Section 3.3 empirically established that RSP reshapes the early-stage output distribution. We now examine whether this shift translates into reasoning diversity at three complementary levels of analysis: token rankings, semantic content of trajectories, and task-level outcomes. The three analyses converge on a directional picture: RSP’s first-token perturbation propagates into semantically more diverse reasoning trajectories, which in turn yields greater output diversity at the task level. All three analyses share the setting: Qwen2.5-Math-1.5B-Instruct, MATH-500, suffix injection, and $L = 10$.

Token-level: distribution beyond temperature. We test whether RSP modifies the first-token distribution in a qualitatively different way from temperature scaling, which preserves token rankings while only flattening probabilities. Under autoregressive decoding, any trajectory-level divergence between RSP and the baseline must originate from the first generation step, a critical forking point for reasoning trajectories [Wang et al., 2025]. Using the baseline at $\tau=1.0$ as reference, we compare RSP at $\tau=1.0$ against the baseline at $\tau=2.0$

(the high-temperature ceiling of what flattening alone can achieve) on three complementary metrics: Spearman rank correlation ρ to detect rank reorganization, the probability mass placed outside the baseline’s top-10 to measure support expansion, and Jensen–Shannon divergence to quantify the overall magnitude of distributional change (formal definitions in Appendix B).

Table 3 reports the contrast. Even at $\tau=2.0$, only 5.15% of probability mass falls outside the baseline’s top-10. RSP at $\tau=1.0$ reaches 27.64%. The Spearman rank correlation tells the story: because temperature scaling $p_i^{(\tau)} \propto p_i^{1/\tau}$ is a monotone transform that preserves token ranking ($p_i > p_j \iff p_i^{(\tau)} > p_j^{(\tau)}$) for any $\tau > 0$, ρ relative to the baseline is exactly 1. RSP, by contrast, reranks the token distribution itself, reducing ρ to 0.651—an effect temperature cannot produce at any τ . The Jensen–Shannon divergence (0.231) quantifies the overall magnitude of this distributional change. Since $\rho < 1$ rules out rank-preserving flattening as the sole cause, the high divergence reflects a structural reshape of the distribution rather than a uniform flattening of the same ranking.

Table 3: First-token distribution metrics measured against the baseline at $\tau=1.0$, comparing the baseline at $\tau=2.0$ with RSP at $\tau=1.0$.

Metric	$\tau=2.0$	RSP
Spearman ρ	1.0	0.651
Mass outside top-10	5.15%	27.64%
JS divergence	0.086	0.231

Trajectory-level: semantic diversity. Token-level reranking raises a second question: do first-token perturbations produce semantically diverse reasoning trajectories, or do independently perturbed trajectories converge to semantically similar ones? We randomly sample 64 problems from MATH-500 and generate 300 independent trajectories per problem under Baseline and RSP at temperature $\tau = 1.0$. Each trajectory is encoded into a dense semantic vector by BGE-M3 [Chen et al., 2024], a sentence-level multilingual embedding model, and we compute three complementary metrics over these embeddings: *pairwise cosine distance* between trajectories captures the overall semantic spread of the trajectory set; *inter-cluster distance* between centroids identified by DBSCAN [Ester et al., 1996], a density-based clustering algorithm that groups nearby trajectories without prespecifying the number of clusters, captures the separation between distinct trajectory groups; and *intra-cluster distance* captures semantic coherence within each group.

Table 4 reveals three concurrent patterns. Pairwise cosine distance rises by about $1.4\times$, indicating that the trajectory set as a whole becomes more semantically diverse. Inter-cluster distance jumps by about $5.4\times$, showing that the trajectories split into substantially more separated groups. Intra-cluster distance is essentially unchanged, indicating that semantic coherence within each group remains comparable to the baseline. RSP also yields larger pairwise distance than baseline on 56 of 64 problems (87.5%).

Table 4: Semantic diversity metrics (64 problems, 300 trajectories per condition).

Metric	Baseline	RSP
Pairwise cos. dist.	0.0612	0.0879
Inter-cluster dist.	0.0183	0.0982
Intra-cluster dist.	0.0526	0.0528

Outcome-level: Pass@ n scaling. Finally, we evaluate whether the token- and trajectory-level diversity translates into task-level outcome diversity through Pass@ n [Chen et al., 2021], the probability that at least one of n sampled solutions is correct. We compare four conditions with $n = 16$ samples per problem on MATH-500 and AIME24: (i) *Baseline*, $\tau=1.0$: temperature sampling only;

198 (ii) *RSP (fixed seed)*, $\tau = 1.0$: a single RSP shared across all samples; (iii) *RSP (indep. seed)*, greedy:
 199 a different RSP per sample without temperature; and (iv) *RSP (indep. seed)*, $\tau = 1.0$: a different RSP
 200 per sample combined with temperature sampling.

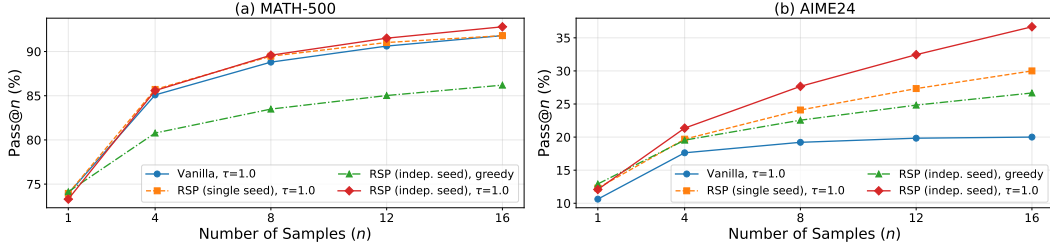


Figure 2: Pass@n scaling on (a) MATH-500 and (b) AIME24 with Qwen2.5-Math-1.5B-Instruct, $n = 16$ samples per problem. Baseline: temperature sampling only. Fixed seed: single RSP shared across samples combined with temperature. Indep. seed: a different RSP per sample, with or without temperature.

201 Condition (iv) achieves the highest Pass@n on both benchmarks, reaching 92.80% on MATH-500 and
 202 36.67% on AIME24 at $n = 16$. Condition (ii) shows no improvement over the baseline, suggesting
 203 that the diversity gain likely depends on varying the random embeddings across samples rather
 204 than a single fixed perturbation. Pass@1 remains comparable to the baseline on MATH-500, and
 205 on the more challenging AIME24, (iv) achieves higher Pass@1 than the baseline. Together these
 206 observations suggest that combining independent RSPs with temperature sampling may yield greater
 207 trajectory diversity while preserving single-sample accuracy.

208 4 Theoretical Analysis: Why RSP Works

209 Our theoretical goal is narrower than a theorem about the full nonlinear transformer. We want
 210 a mechanism that explains the three empirical regularities from Section 3: random continuous
 211 prompts can alter the model’s earliest reasoning decisions even though they are not optimized for
 212 the task; their influence is strongest at the beginning of decoding and weakens later; and this early
 213 perturbation can increase trajectory diversity without permanently destabilizing generation. The
 214 common control variable throughout the section is $w_{r,i}^{(l)}$, the attention mass assigned to the random
 215 tokens. The argument therefore has three parts. First, we characterize what kind of perturbation can
 216 open reasoning branches without inserting a hand-crafted prior, and explain why centered isotropic
 217 randomness is the relevant choice. Second, we show that attention injects the perturbation through
 218 $w_{r,i}^{(l)}$ and simultaneously places an annealing pressure on it as real-token evidence grows. Third, we
 219 show why this pre-softmax perturbation can change token rankings in a way that temperature scaling
 220 cannot.

221 4.1 Why Random Rather Than a Fixed Token?

222 A generic exploration prompt for reasoning should satisfy three structural requirements. It should
 223 not systematically pre-commit the model to one semantic direction before the input-dependent
 224 computation begins. It should vary across trials, otherwise repeated decoding would trace the same
 225 perturbation every time. And because the branch-sensitive direction is unknown in advance and
 226 depends on the prompt, the decoding state, and the token comparison, the perturbation should not
 227 privilege some directions while leaving others almost untouched. Optimized soft prompts deliberately
 228 encode a task prior [Kang et al., 2026, Xu et al., 2025, Ye et al., 2026, Zhang et al., 2026]; our
 229 question is the complementary one: what perturbation should one use when no such prior is assumed?

230 A fixed token or a fixed continuous prompt fails this requirement set immediately. It may steer one
 231 trajectory, but if the same vector is reused across runs then there is no across-trial exploration at all.
 232 Resampling discrete vocabulary tokens restores variability, yet each draw still comes with its own
 233 lexical direction and there is no reason for the resulting perturbations to be centered or isotropic. We
 234 formalize the choice we make instead.

Definition 1 (Random Soft Prompt). Let $\mathbf{W}_E \in \mathbb{R}^{V \times d}$ denote the pretrained input-embedding table, with μ_E and σ_E the entrywise scalar mean and standard deviation defined in Section 3.1. An RSP of length L is a sequence $\mathbf{H} = (h_1, \dots, h_L)$ with

$$h_l = \mu_E \mathbf{1}_d + \sigma_E \xi_l, \quad \xi_l \stackrel{iid}{\sim} \mathcal{N}(0, \mathbf{I}_d),$$

matching the implementation in Section 3.1 ($h_l \sim \mathcal{N}(\mu_E, \sigma_E^2 \mathbf{I}_d)$, with μ_E broadcast as $\mu_E \mathbf{1}_d$). Equivalently, the centered prompt $\bar{h}_l := h_l - \mu_E \mathbf{1}_d$ satisfies $\bar{h}_l \sim \mathcal{N}(0, \sigma_E^2 \mathbf{I}_d)$. Whenever we later analyze repeated-sampling protocols such as Pass@N, we additionally assume that the prompt draws and any decoding randomness used across compared trials are independent.

The proofs in this section use only the centered law of $\bar{h}_l$, not the particular form of the centering vector $\mu_E \mathbf{1}_d$. The two embedding-derived statistics from Section 3.1 play distinct roles in this design. The scalar σ_E^2 borrows the *scale* of the pretrained embedding distribution so that the perturbation operates in the magnitude regime the model expects. The isotropic covariance $\sigma_E^2 \mathbf{I}_d$, in contrast, deliberately discards the directional structure of the embedding distribution; Theorem 1 below shows that this is the unique covariance that maximizes the worst-case directional variance under a fixed trace budget. The downstream attention mechanism then maps this prompt-space distribution to value space via the value projection $\mathbf{W}_V^{(l)}$; the corresponding variance facts are stated explicitly below.

Lemma 1 (Semantic neutrality). Let $\bar{h} \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_d)$ and $\hat{h} := \bar{h} / \|\bar{h}\|$. For any fixed unit vectors $u_1, \dots, u_M \in \mathbb{S}^{d-1}$ and any $\delta \in (0, 1)$,

$$\Pr \left[\max_{r \in [M]} |\hat{h}^\top u_r| \leq \sqrt{\frac{2 \log(2M/\delta)}{d-1}} \right] \geq 1 - \delta.$$

In contrast, for any nonzero token embedding e_s , choosing the anchor direction $u = \hat{e}_s := e_s / \|e_s\|$ yields $|\hat{e}_s^\top u| = 1$.

Lemma 1 gives the precise sense in which the perturbation is neutral. The claim is not that a random vector is literally free of every incidental semantic correlation. The relevant and weaker statement is that it does not systematically align with any pre-selected finite family of semantic alternatives. Centering then removes deterministic first-order drift: under the local surrogate below, the expected perturbation of any logit gap is zero when $\mathbb{E}[\bar{h}] = 0$.

To formalize the exploration part, we specialize to the per-token surrogate that treats the contribution of a single random token at a time. Let $\bar{h} \in \mathbb{R}^d$ denote one centered prompt vector $\bar{h}_l$ from Definition 1. The multi-token attention output $\eta_i^{(l)} = \sum_{j=1}^k \alpha_{i,n+j}^{(l)} \mathbf{v}_{r_j}^{(l)}$ analyzed in §4.2 aggregates L such independent draws through softmax-weighted attention, where the weights $\alpha_{i,n+j}^{(l)}$ themselves depend on the random tokens; the per-token surrogate identifies how a single $\bar{h}_j$ influences the logit gap, and the multi-token case is empirically captured by Figure 3 and the Pass@N analysis of Section 3.4. With this convention, consider the local first-order surrogate

$$z_i(\bar{h}) \approx z_i + a_i^\top \bar{h}, \quad a_i := \nabla_{\bar{h}} z_i(0),$$

valid for sufficiently small $\|\bar{h}\|$ (Appendix D); we use this surrogate throughout the section to identify the direction of perturbation rather than to predict its exact magnitude. For a token pair (i, j) , the perturbed logit gap becomes

$$\Delta_{ij}(\bar{h}) \approx \Delta_{ij} + b_{ij}^\top \bar{h}, \quad b_{ij} := a_i - a_j.$$

The relevant question is therefore not “does the perturbation have variance?” but “how much variance does it retain along the unknown direction b_{ij} that matters for this branching decision?” The theorem below answers that question at the level of second moments.

Theorem 1 (Direction-agnostic coverage). Let D be any zero-mean perturbation distribution on $\mathbb{R}^d$ with covariance Σ_D and variance budget $\text{tr}(\Sigma_D) \leq \rho^2$. Define the worst directional variance

$$\mathcal{E}(D) := \min_{\|b\|=1} \text{Var}_{h \sim D}(b^\top h).$$

Then

$$\mathcal{E}(D) = \lambda_{\min}(\Sigma_D) \leq \frac{\rho^2}{d},$$

with equality if and only if $\Sigma_D = (\rho^2/d) \mathbf{I}_d$. Hence isotropic covariance is the unique way to equalize first-order perturbation variance across all directions while maximizing the least-covered one.

Theorem 1 is not meant to cast prompting as an adversarial game; its role is simpler. When the branch-sensitive direction is unknown at sample time, isotropy is the unique second-order description that neither favors nor neglects any direction. A fixed prompt has $\Sigma_D = 0$ and therefore creates no across-trial exploration at all. An anisotropic perturbation leaves at least one direction less explored than the isotropic optimum. The Gaussian choice in RSP is the simplest centered, rotation-symmetric family that realizes this covariance while matching the scale of pretrained embeddings. This is the sense in which RSP is not just “some noise”: it is a resampled, centered, direction-agnostic perturbation.

A natural alternative one might consider is to draw the perturbation from a distribution that *matches* the empirical row-covariance geometry of the pretrained embedding table $\mathbf{W}_E$, rather than discarding it. Theorem 1 dismisses this alternative as soon as that row-covariance is non-isotropic.

Corollary 1 (Embedding-matched noise is suboptimal in worst-direction coverage). *Let $(\mathbf{W}_E)_v \in \mathbb{R}^d$ denote the v -th row of $\mathbf{W}_E$ and define the row-covariance of $\mathbf{W}_E$ around the same scalar-broadcast center $\mu_E \mathbf{1}_d$ used in Definition 1,*

$$\Sigma_E := \frac{1}{V} \sum_{v=1}^V ((\mathbf{W}_E)_v - \mu_E \mathbf{1}_d)((\mathbf{W}_E)_v - \mu_E \mathbf{1}_d)^\top, \quad \text{tr}(\Sigma_E) = d \sigma_E^2.$$

Consider two zero-mean perturbations on $\mathbb{R}^d$ with the same trace budget $\rho^2 := d \sigma_E^2$: the isotropic RSP law D_{iso} of Definition 1 with covariance $\sigma_E^2 \mathbf{I}_d = (\rho^2/d) \mathbf{I}_d$, and the embedding-matched law D_{emb} with covariance Σ_E . Then

$$\mathcal{E}(D_{\text{iso}}) = \frac{\rho^2}{d} \geq \lambda_{\min}(\Sigma_E) = \mathcal{E}(D_{\text{emb}}),$$

with strict inequality whenever Σ_E is non-isotropic.

The corollary is one application of Theorem 1 and requires no separate proof; the trace identity $\text{tr}(\Sigma_E) = d \sigma_E^2$ follows directly from the same scalar centering convention used in Section 3.1. The motivation for considering this alternative comes from a line of work documenting that pretrained word- and sentence-embedding representations tend to be anisotropic rather than uniformly distributed in $\mathbb{R}^d$, with a dominant common direction and a few high-variance directions [Mu and Viswanath, 2018] and a more general anisotropy in contextualized representations [Ethayarajh, 2019]. To the extent that the row-covariance Σ_E inherits any such anisotropy, $\lambda_{\min}(\Sigma_E)$ is strictly smaller than ρ^2/d , so embedding-matched noise leaves at least one direction less explored than the isotropic comparator. Discarding the directional geometry of $\mathbf{W}_E$ while keeping its scale is therefore not a simplification of an embedding-matched alternative but a deliberate choice for direction-agnostic exploration.

Lemma 1 and Theorem 1 therefore play complementary roles. The lemma rules out semantic pre-commitment to any fixed anchor set, while the theorem rules out directional under-coverage in the local branch-sensitive geometry. A perturbation that satisfies only the first condition would be neutral but too weak in some directions; a perturbation that satisfies only the second would still inject a hand-chosen semantic prior. RSP is useful because it satisfies both at once.

4.2 How Attention Turns a Neutral Perturbation into Controlled Exploration

The previous subsection explains why centered isotropic randomness is the right kind of perturbation. The next question is temporal: why is the perturbation strong enough to matter at the beginning of decoding but weak enough not to derail the rest of the reasoning chain? The formal answer uses three ingredients. First, in an idealized Hopfield-style view of attention, each transformer block acts like one step of an iterative attractor process; in that picture the unperturbed path is strongly constrained by its initialization, which identifies a structural source of exploration deficit. Second, the attention output admits an exact decomposition into a real-token update and a random-token contribution. Third, the same attention mass that injects the random contribution is structurally forced downward as real-token evidence accumulates, and the perturbation propagates through the feed-forward sublayer with controlled amplification.

Iterative attractor view. Following the Modern Hopfield correspondence [Ramsauer et al., 2021], one attention update with key matrix $\mathbf{K}$ and value matrix $\mathbf{V}$ is equivalent to a gradient step on the

324 energy

$$E_{\mathbf{K}}(\sigma) = -\beta^{-1} \log \sum_j \exp(\beta \mathbf{k}_j^\top \sigma) + \frac{1}{2} \|\sigma\|^2$$

325 when $\mathbf{K} = \mathbf{V}$. Under three idealized assumptions (A1) $\mathbf{K} = \mathbf{V}$, (A2) *weight-tied attention*, (A3)
 326 *L_E -smoothness of $E_{\mathbf{K}}$ with $L_E < 2$* , Appendix D.1 shows that the unperturbed iterates produce a
 327 strictly monotone descent in $E_{\mathbf{K}}$ and the sublevel set $\{\sigma : E_{\mathbf{K}}(\sigma) \leq E_{\mathbf{K}}(\sigma^{(0)})\}$ is forward-invariant;
 328 under the additional separation condition $\text{sep}(C(\sigma^{(0)})) > 0$ together with the per-step gradient bound
 329 of Corollary 3, the trajectory also remains in the connected component containing $\sigma^{(0)}$.

330 (A1) and (A2) are not literally satisfied by real transformers, so the Hopfield view here is an idealized
 331 structural picture rather than a literal statement about every transformer block. Its role is to identify
 332 why a deterministic forward pass can be exploration-poor: under A1–A3 it stays inside a fixed
 333 sublevel set, and under the extra separation/small-step condition it also remains in the connected
 334 component selected by initialization. We use this picture to motivate why injecting an isotropic
 335 stochastic perturbation at every block is the relevant intervention, not to claim formal escape rates for
 336 real networks.

337 Throughout the rest of the section we work with the actual transformer attention. *At the first attention*
 338 *block* ($l = 1$), the random tokens have not yet been transformed, so the random values are obtained
 339 by applying the value projection directly to the centered prompt vectors:

$$\mathbf{v}_{r_j}^{(1)} = \mathbf{W}_V^{(1)} \bar{h}_j, \quad \text{Cov}(\mathbf{v}_{r_j}^{(1)}) = \sigma_E^2 \mathbf{W}_V^{(1)} \mathbf{W}_V^{(1)\top}.$$

340 Proposition 5 in Appendix C states the resulting directional-variance bounds explicitly: if $\mathbf{W}_V^{(1)}$
 341 is full rank the projected perturbation has strictly positive variance in every value-space direction,
 342 and if $\mathbf{W}_V^{(1)}$ is well-conditioned the prompt-space direction-agnostic coverage transfers to first-layer
 343 value space up to layer-dependent constants. At deeper layers $l > 1$, the random tokens evolve
 344 through prior attention/FFN updates, so the analogous covariance pushforward applies to the *current*
 345 *random-token state* $\mathbf{x}_{r_j}^{(l)}$ rather than to $\bar{h}_j$ directly. The remaining quantitative results in this subsection
 346 (decomposition, w_r decay, FFN propagation) do not require the deep-layer covariance bound; they
 347 use only the algebraic decomposition and Lipschitz control. We therefore use first-layer directional
 348 coverage as the entry-point guarantee and rely on the propagation bounds below for cross-block
 349 control.

350 **Exact decomposition.** When k random vectors are inserted as additional keys and values, the
 351 attention update can be written as

$$\mathbf{x}_i^{(l+1)} = (1 - w_{r,i}^{(l)}) \tilde{\mathbf{x}}_i^{(l+1)} + \boldsymbol{\eta}_i^{(l)}, \quad (2)$$

352 where $\tilde{\mathbf{x}}_i^{(l+1)}$ is the update produced by the real tokens alone, $\boldsymbol{\eta}_i^{(l)} := \sum_{j=1}^k \alpha_{i,n+j}^{(l)} \mathbf{v}_{r_j}^{(l)}$ is the
 353 contribution of the random values, and

$$w_{r,i}^{(l)} := \sum_{j=1}^k \alpha_{i,n+j}^{(l)}$$

354 is the total attention mass assigned to the random tokens. Appendix D.2 proves the decomposition
 355 and the deterministic bound

$$\|\boldsymbol{\eta}_i^{(l)}\| \leq w_{r,i}^{(l)} \max_{j \in [k]} \|\mathbf{v}_{r_j}^{(l)}\|.$$

356 This is the structural core of the mechanism: the same scalar $w_{r,i}^{(l)}$ controls how much of the original
 357 update is retained and how large the random contribution can be. If $w_{r,i}^{(l)}$ is large, the perturbation can
 358 materially change the update; if $w_{r,i}^{(l)}$ is small, the computation collapses back toward the real-token
 359 update.

360 **Natural annealing through attention.** The second structural fact is that $w_{r,i}^{(l)}$ is not arbitrary. It
 361 satisfies

$$w_{r,i}^{(l)} \leq \frac{k}{k + n \cdot \exp(\Delta_i^{(l)} / \sqrt{d})}, \quad (3)$$

where $\Delta_i^{(l)} := \min_{j \in [n]} \mathbf{q}_i^\top \mathbf{k}_j - \max_{j' \in [k]} \mathbf{q}_i^\top \mathbf{k}_{r_{j'}}$, is the real-vs-random logit gap and n is the number of real tokens. This bound yields a clean *sequence-wise dilution effect*: for a fixed gap, every newly generated real token tightens the maximum random-token attention compatible with that state. A similar *layer-wise* decay is expected whenever deeper layers enlarge the real-vs-random gap $\Delta_i^{(l)}$. We intentionally treat that layer-wise statement as a model-side hypothesis rather than as a universal theorem, because it depends on how the model sharpens real-token retrieval with depth. The resulting prediction is simple: the perturbation should be strongest exactly when the model has the least context and many continuations remain open, and it should weaken as evidence accumulates.

Theorem 2 (Sequence-wise annealing under a fixed logit gap). *Fix k and $\Delta_i^{(l)}$. The upper bound in Eq. (3) is strictly decreasing in the number n of real tokens. Consequently, cache growth alone imposes a monotone decrease on the largest random-token attention compatible with that logit gap.*

Theorem 2 proves a structural dilution effect, not monotone decay of the realized $w_{r,i}^{(l)}$ on every decoding path, because $\Delta_i^{(l)}$ also evolves with state. The layer-wise half is likewise model-specific and must be checked empirically.

Feed-forward propagation under local Lipschitz control. The decomposition above governs the attention sublayer. A real transformer block applies a feed-forward sublayer $F^{(l)}$ after attention with residual connections, so the cross-block perturbation depends on how $F^{(l)}$ propagates $\boldsymbol{\eta}_i^{(l)}$. If $F^{(l)}$ is $L^{(l)}$ -Lipschitz on a bounded set containing both the perturbed and unperturbed post-attention states, then Appendix D.4 shows

$$\|\mathbf{x}_i^{(l+1, \text{RSP})} - \mathbf{x}_i^{(l+1, \text{base})}\| \leq (1 + L^{(l)}) (\|\boldsymbol{\eta}_i^{(l)}\| + w_{r,i}^{(l)} \|\bar{\mathbf{x}}_i^{(l+1)}\|),$$

which together with $\|\boldsymbol{\eta}_i^{(l)}\| \leq w_{r,i}^{(l)} \max_j \|\mathbf{v}_{r_j}^{(l)}\|$ shows that the per-block perturbation amplitude scales linearly with $w_{r,i}^{(l)}$. The bounded-set assumption is plausible in practice because LayerNorm constrains the post-attention norm, so the perturbed and unperturbed states traversed during a single forward pass lie in a common bounded region. Iterating across the fixed finite stack yields the explicit cumulative bound of Corollary 7. Under standard architectural assumptions this bound is finite for a fixed model depth, and the later-block injections themselves shrink as $w_{r,i}^{(l)}$ decreases. Stronger uniform-in-depth non-expansion would require additional contraction assumptions, which we do not invoke here.

Empirical verification. We test exactly these model-dependent steps on Qwen2.5-Math-7B under suffix injection. If $w_{r,i}^{(l)}$ really acts as an annealed exploration coefficient, then the realized attention to RSP tokens should decay across reasoning positions and layers, whereas attention to question tokens should not exhibit the same monotone pattern because they remain task-relevant throughout decoding. Figure 3 shows precisely this contrast. The figure therefore serves a specific logical role: it validates the model-side sharpening behavior that Eq. (3) by itself does not prove.

4.3 Output Distribution Shift and Exploration

The practical consequence of the previous mechanism is branch opening. To connect that mechanism to observable decoding behavior, we now return to the output distribution. The relevant question is not whether RSP merely flattens probabilities, but whether it can make the decoder consider tokens that temperature scaling can never promote into contention.

Why temperature is insufficient. Temperature scaling transforms probabilities as $p_i^{(\tau)} \propto p_i^{1/\tau}$ and preserves token ranking, i.e., $p_i > p_j \iff p_i^{(\tau)} > p_j^{(\tau)}$. It can soften or sharpen confidence, but it cannot create a new ordering among candidate tokens at a fixed decoding state.

Why RSP changes rankings. RSP acts before softmax, through the hidden state. At a fixed decoding state, temperature rescales existing logit gaps, whereas RSP perturbs the gaps themselves. Using the same first-order local surrogate as above, define

$$z_i^{\text{lin}}(\bar{h}) := z_i + a_i^\top \bar{h}.$$

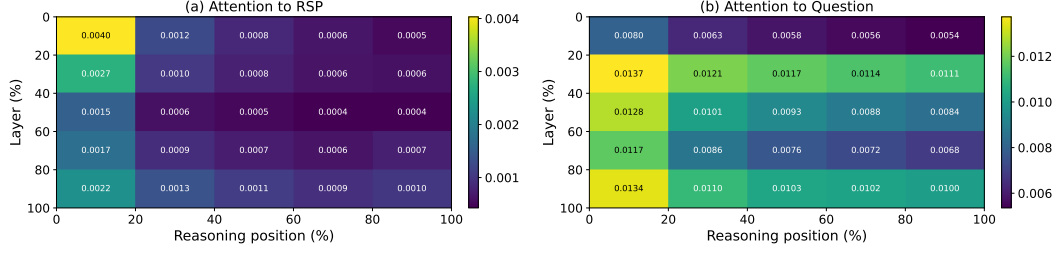


Figure 3: Per-token attention mass on Qwen2.5-Math-7B (suffix, 500 MATH-500 problems, each with an independently sampled RSP). The layer and reasoning-position axes are each partitioned into five quantile bins; every cell reports the 500-sample mean per-token attention mass directed to (a) the RSP embeddings and (b) the question tokens within that region. The horizontal axis spans reasoning position from early (left) to late (right) generation steps, and the vertical axis spans layer depth from shallow (top) to deep (bottom). Attention to the RSP embeddings in (a) decays along both axes, while attention to the question tokens in (b) remains comparatively uniform.

406 The next two propositions are statements about these surrogate logits, not the exact transformer logits;
 407 this scope is sufficient for the comparison to temperature because temperature is rank-preserving
 408 regardless of approximation.

409 The next two propositions use only one additional support property of the centered prompt law: every
 410 nonempty open subset of $\mathbb{R}^d$ receives positive probability. The Gaussian instantiation of Definition 1
 411 satisfies this automatically, but the proofs themselves do not require Gaussian tails once this support
 412 condition is in place.

413 **Proposition 1** (Stochastic ranking under the local affine surrogate). *If there exists a token pair (i, j)*
 414 *such that $a_i \neq a_j$, and if the centered prompt law assigns positive probability to every nonempty*
 415 *open subset of $\mathbb{R}^d$, then*

$$0 < \Pr_{\bar{h}} (z_i^{\text{lin}}(\bar{h}) > z_j^{\text{lin}}(\bar{h})) < 1.$$

416 *Thus, unlike temperature scaling, the local affine surrogate of RSP randomizes token rankings at each*
 417 *decoding step.*

418 Proposition 1 is the minimal statement: pairwise order can flip. Exploration requires a stronger
 419 conclusion, namely that tokens outside the baseline candidate set can actually enter it.

420 **Proposition 2** (Top- K entry under a nonempty surrogate entry region). *Let $J_K(s_t)$ be the baseline*
 421 *top- K set at decoding state s_t , and let $i \notin J_K(s_t)$. Define*

$$\mathcal{G}_{i,K} := \{\bar{h} \in \mathbb{R}^d : \# \{j \neq i : z_j^{\text{lin}}(\bar{h}) > z_i^{\text{lin}}(\bar{h})\} \leq K - 1\}.$$

422 *If $\mathcal{G}_{i,K}$ contains a nonempty open set and the centered prompt law assigns positive probability to*
 423 *every nonempty open subset of $\mathbb{R}^d$, then*

$$\Pr_{\bar{h}} (\bar{h} \in \mathcal{G}_{i,K}) > 0.$$

424 Proposition 2 is intentionally conditional: it does not claim that every excluded token admits such a
 425 top- K entry region. It states that whenever the first-order model contains an open region in prompt
 426 space where a previously excluded token moves into the top- K , the centered perturbation reaches
 427 that region with positive probability. A sufficient condition for the open-region hypothesis is the
 428 following *strict-margin admission* condition. For each subset $S \subseteq \{j \neq i\}$ with $|S| \leq K - 1$, define
 429 the strict-inequality polyhedron

$$\mathcal{O}_S := \left\{ \bar{h} \in \mathbb{R}^d : z_j + a_j^\top \bar{h} > z_i + a_i^\top \bar{h} \forall j \in S, \quad z_i + a_i^\top \bar{h} > z_{j'} + a_{j'}^\top \bar{h} \forall j' \notin S \cup \{i\} \right\}.$$

430 Each $\mathcal{O}_S$ is an open subset of $\mathbb{R}^d$ as a finite intersection of open half-spaces, and any $\bar{h} \in \mathcal{O}_S$ has
 431 token i ranked $|S| + 1 \leq K$, hence $\bar{h} \in \mathcal{G}_{i,K}$. Therefore $\bigcup_{|S| \leq K-1} \mathcal{O}_S$ is an open subset of $\mathcal{G}_{i,K}$,
 432 and $\mathcal{G}_{i,K}$ contains a nonempty open set whenever some $\mathcal{O}_S$ is nonempty. Equivalently, the existence
 433 of any $\bar{h}_0 \in \mathbb{R}^d$ at which token i outranks all but at most $K - 1$ other tokens with a strict margin

434 yields $\bar{h}_0 \in \mathcal{O}_S$ for $S = \{j : z_j + a_j^\top \bar{h}_0 > z_i + a_i^\top \bar{h}_0\}$, and the open polyhedron $\mathcal{O}_S$ contains an
 435 open ball around $\bar{h}_0$. The inclusion $\bigcup_S \mathcal{O}_S \subseteq \mathcal{G}_{i,K}$ is one-way (ties leave a measure-zero subset of
 436 $\mathcal{G}_{i,K}$ outside $\bigcup_S \mathcal{O}_S$), but it is sufficient for Proposition 2 since only nonempty open subsets of $\mathcal{G}_{i,K}$
 437 are needed. The complementary case—no such $\bar{h}_0$ exists—is exactly the situation in which token i
 438 is dominated by the baseline top- K in the affine logit surrogate, and Proposition 2 makes no claim
 439 there. Section 3.4 reports the probability mass that RSP places outside the baseline top-10 at the first
 440 generation step. This mass-outside-top- K statistic is a coarse aggregate rather than a direct count
 441 of top- K admission events; it is consistent with the existence of nonempty entry regions for at least
 442 some decoding states without formally estimating the surrogate open-set condition. Gaussianity is
 443 one sufficient way to realize the required support condition; temperature cannot reach $\mathcal{G}_{i,K}$ because
 444 it never changes token order at all. The mechanism here is branch opening, not mere probability
 445 flattening.

446 **Corollary 2** (Sample complexity of top- K admission). *Let $\mu_i := \Pr_{\bar{h}}(\bar{h} \in \mathcal{G}_{i,K}) > 0$ denote the*
 447 *per-sample admission probability for token i . Within N independent fresh RSP draws, the probability*
 448 *of at least one admission is at least*

$$1 - (1 - \mu_i)^N \geq 1 - \exp(-N\mu_i).$$

449 Corollary 2 closes the loop from per-sample positive probability to multi-trial exploration. Even
 450 when μ_i is small, $N \cdot \mu_i = \Theta(1)$ samples suffice to admit a previously excluded token with constant
 451 probability. The independence assumption first enters here: the single-run rank-flip and top- K entry
 452 statements above do not require it. This is the formal bridge from the static guarantee of Proposition 2
 453 to the trajectory-level Pass@ N phenomena reported in Section 3.4.

454 **From rank flips to trajectory diversity to accuracy gain.** Together, Propositions 1 and 2 close the
 455 logical chain from Section 3.3 to Section 3.4, up to the local-surrogate assumption above. Early RSP
 456 perturbations alter logit gaps before softmax, which can admit previously excluded tokens into the
 457 candidate set. Because autoregressive decoding compounds early choices, one such early admission is
 458 enough to push the trajectory into a different reasoning cluster even if later perturbations are already
 459 small. RSP therefore does not need to perturb the entire chain of thought uniformly: it only needs to
 460 alter the first few branching decisions before attention-mediated attenuation suppresses further noise,
 461 which is why early exploration and late stabilization coexist in the same mechanism.

462 The connection from trajectory diversity to accuracy gain requires a property of reasoning tasks
 463 themselves: the existence of multiple valid trajectories. We formalize this as a separate proposition
 464 rather than absorb it into the perturbation analysis.

465 **Proposition 3** (Exploration-meets-multi-path accuracy gain). *Fix a problem x , and consider the pro-*
 466 *cedure that runs the deterministic baseline once and additionally draws N fresh RSPs $\bar{h}^{(1)}, \dots, \bar{h}^{(N)}$,*
 467 *with the model decoded once per sample. Let $\mathcal{T}^*(x)$ denote the (nonempty) set of trajectories*
 468 *reaching the correct answer and $p_{\text{base}}(x) \in \{0, 1\}$ the deterministic-baseline correctness indicator.*
 469 *Assume (M1) for each $n \in \{1, \dots, N\}$, the marginal probability that the n -th RSP run produces a*
 470 *trajectory in $\mathcal{T}^*(x)$ is at least $p_{\min}(x) > 0$ (the same lower bound for every n), and (M2) the N RSP*
 471 *runs are mutually independent (independent prompt draws and, if decoding is stochastic, independent*
 472 *decoding randomness across runs). Then*

$$\Pr(\text{Pass}@ (N+1) \text{ correct on } x) \geq \max\{p_{\text{base}}(x), 1 - (1 - p_{\min}(x))^N\} \geq 1 - e^{-Np_{\min}(x)}.$$

473 *The expected accuracy gain over the deterministic baseline is at least*

$$\frac{1}{|\mathcal{D}|} \sum_{x \in \mathcal{D}} (1 - p_{\text{base}}(x)) (1 - \exp(-Np_{\min}(x))).$$

474 *Proof.* Let $p_n(x) \geq p_{\min}(x)$ denote the per-run success probability of the n -th RSP run as in (M1).
 475 The Pass@ $(N+1)$ event holds if the baseline run reaches $\mathcal{T}^*(x)$ (probability $p_{\text{base}}(x) \in \{0, 1\}$) or
 476 if any of the N RSP runs does. By (M2), the failure probabilities of the N RSP runs multiply, so
 477 the probability that at least one RSP run reaches $\mathcal{T}^*(x)$ equals $1 - \prod_{n=1}^N (1 - p_n(x))$, which is at
 478 least $1 - (1 - p_{\min}(x))^N$ by (M1). Taking the maximum with $p_{\text{base}}(x)$ gives the first lower bound;
 479 $1 - x \leq e^{-x}$ gives the second. Restricting to $\{x : p_{\text{base}}(x) = 0\}$ yields the expected gain. $\square$

Proposition 3 converts the local guarantees of Propositions 1 and 2 into a global accuracy claim, but only via the auxiliary assumption that $p_{\min}(x) > 0$. The next lemma formalizes a sufficient condition under which the perturbation theory of this section actually delivers $p_{\min}(x) > 0$, separating what is proved (positive measure of an entry region) from what is assumed (extension of a perturbed prefix to a correct completion).

Lemma 2 (Bridge from open entry region to one-run success). *Specialize Proposition 2 to $K = 1$ (greedy decoding) and assume the same support condition on the centered prompt law. Suppose there exists a nonempty open ball $B \subseteq \mathbb{R}^d$ such that for every $\bar{h} \in B$ both of the following hold under a single RSP draw at the start of decoding: (B1) the RSP-induced trajectory under prompt $\bar{h}$ reaches some decoding state $s_{t^*}(\bar{h})$ at which an excluded token $i^*(\bar{h}) \notin J_1(s_{t^*})$ satisfies $\bar{h} \in \mathcal{G}_{i^*,1}(s_{t^*})$, so greedy decoding selects $i^*(\bar{h})$ at step t^* ; (B2) the resulting full trajectory belongs to $\mathcal{T}^*(x)$. Then $p_{\min}(x) \geq \Pr(\bar{h} \in B) > 0$.*

Proof. By definition, $p_{\min}(x)$ is the per-sample probability that the RSP-induced full trajectory under one draw of $\bar{h}$ reaches $\mathcal{T}^*(x)$. Conditions (B1) and (B2) state exactly that this event holds for every $\bar{h} \in B$. Since B is a nonempty open subset of $\mathbb{R}^d$ and the centered prompt law assigns positive probability to every nonempty open set, $\Pr(\bar{h} \in B) > 0$, and the lower bound follows. $\square$

Lemma 2 converts the abstract assumption $p_{\min}(x) > 0$ of Proposition 3 into the existence of an open ball B satisfying two simultaneous properties: (B1) the perturbation theory of this section—specifically Proposition 2 at $K = 1$ —supplies the open-set structure that triggers top-1 entry once a candidate decoding state s_{t^*} is fixed; (B2) the resulting full trajectory continues into a correct completion. The prefix-preservation aspect (the trajectory under $\bar{h}$ actually reaches a state s_{t^*} with the desired entry-region structure) and the completion property (B2) are both model- and dataset-dependent and treated as empirical hypotheses. Section 3.4 plays a load-bearing role here: it verifies that RSP-induced trajectories include both improvements and regressions over the baseline with net gain emerging under repeated sampling, which is precisely the empirical signature predicted by such an open ball B existing for some but not all problems. The mechanism we have isolated is exploration meeting multi-path structure, not steering toward correctness.

Taken together, the causal chain is now explicit. Randomness matters because RSP is resampled, centered, and direction-agnostic; attention turns that perturbation into an early exploration coefficient and then anneals it; the resulting early rank flips open trajectories that temperature scaling alone cannot reach; and the multi-path structure of reasoning converts trajectory diversity into accuracy gain under repeated sampling. Section 3 therefore reports four observable faces of a single mechanism rather than four disconnected phenomena.

5 Conclusion

Summary. This work investigated the role of Random Soft Prompts (RSPs), continuous embeddings sampled from the pretrained token embedding distribution and injected into the model input without any training. Through empirical and theoretical analyses, we established three findings. First, RSPs increase early-stage token entropy, top-1 probability reduction, and varentropy while the model’s output stabilizes in later generation stages (Section 3). Second, RSPs induce stochastic token rankings and expand the effective support of the output distribution, a property that temperature sampling fundamentally lacks due to its ranking-preserving nature (Section 4). Third, combining RSPs with temperature sampling produces more diverse reasoning trajectories than either method alone, as measured by Pass@ n scaling (Section 3).

Implications for GRPO training. GRPO computes advantages by comparing multiple sampled trajectories within each group, making trajectory diversity essential for effective learning signals. Existing approaches such as DAPO [Yu et al., 2025] and CE-GPPO [Su et al., 2026] address entropy collapse by modifying the clipping range or reintroducing gradients from clipped tokens, but the underlying sampling process remains based on temperature sampling, which preserves token rankings and cannot expand the output support. Our theoretical and empirical results suggest that RSP can address this limitation at the sampling level: by perturbing the initial output distribution with independent random embeddings per rollout, RSP provides a complementary source of diversity

that is orthogonal to temperature-based exploration. We expect this to yield more effective learning signals for GRPO training, particularly in mitigating entropy collapse.

Limitations. Our experiments are conducted on mathematical reasoning benchmarks (MATH-500, GSM8K, AIME24) with a limited set of model families (Qwen2.5-Math, LLaMA-3.1). The energy-theoretic analysis in Section 4 relies on strong assumptions: the Hopfield–Attention equivalence (A1: $\mathbf{K} = \mathbf{V}$) and weight-tied layers (A2), which hold exactly only in Hopfield-type models and deep equilibrium models, respectively. Since real transformer architectures vary widely in their attention mechanisms and layer structures, the theoretical predictions are most directly applicable to models whose attention dynamics closely approximate the Hopfield energy framework. The GRPO application remains a theoretical expectation and has not been empirically validated.

Future work. Extending the experimental evaluation to other domains (e.g., code generation, commonsense reasoning) and larger model scales would test the generality of RSP’s effects. Empirically validating the GRPO application by training with RSP-augmented rollouts and comparing against DAPO and CE-GPPO is a natural next step. Investigating the optimal RSP distribution and token count, which currently require per-model tuning, could lead to more principled injection strategies.

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 654 Pierre Roux, Piotr Dollar, Polina Zvyagina, Prashant Ratanandani, Pritish Yuvraj, Qian Liang,
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 656 Rangaprabhu Parthasarathy, Raymond Li, Rebekkah Hogan, Robin Battey, Rocky Wang, Russ
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740 A Experimental Setup and Full Accuracy Breakdown

741 Table 5 provides the full per-position accuracy breakdown (Prefix, Infix, Suffix) corresponding to
 742 the best-across-positions results summarized in Table 1 in the main text. Table 6 lists the number of
 743 RSP tokens L used for each model–benchmark pair, selected from $\{10, 15, 20\}$. All experiments are
 744 conducted on a single NVIDIA A6000 40GB GPU with greedy decoding, a maximum generation
 745 length of 3,072 tokens for MATH-500 and GSM8K, and 4,096 tokens for AIME24. Batch size is
 746 fixed per model (16 for LLaMA-3.1-8B-Instruct and Qwen2.5-Math-7B variants, 32 for Qwen2.5-
 747 Math-1.5B variants).

Table 5: Accuracy (%) across model configurations, injection positions, and benchmarks. Values in parentheses indicate the difference from the baseline. **Bold** denotes the best result and underline denotes the second best for each model–benchmark pair.

Model	Mode	MATH-500	GSM8K	AIME24
<i>Instruct Models</i>				
LLaMA-3.1-8B-Instruct	Baseline	52.20	85.75	<u>6.67</u>
	Prefix	45.00 (−7.2)	76.35 (−9.4)	3.33 (−3.3)
	Infix	49.40 (−2.8)	85.97 (+0.2)	10.00 (+3.3)
	Suffix	<u>49.40</u> (−2.8)	86.73 (+1.0)	10.00 (+3.3)
Qwen2.5-Math-7B-Instruct	Baseline	83.20	95.45	<u>13.33</u>
	Prefix	81.40 (−1.8)	94.92 (−0.5)	13.33 (+0.0)
	Infix	84.20 (+1.0)	95.60 (+0.2)	16.67 (+3.3)
	Suffix	<u>83.40</u> (+0.2)	95.68 (+0.2)	16.67 (+3.3)
Qwen2.5-Math-1.5B-Instruct	Baseline	73.00	<u>85.29</u>	10.00
	Prefix	74.80 (+1.8)	85.29 (+0.0)	<u>13.33</u> (+3.3)
	Infix	75.20 (+2.2)	85.75 (+0.5)	6.67 (−3.3)
	Suffix	74.20 (+1.2)	84.69 (−0.6)	20.00 (+10.0)
<i>Base Models (Raw Text)</i>				
Qwen2.5-Math-7B	Baseline	72.20	84.15	6.67
	Prefix	<u>71.60</u> (−0.6)	84.31 (+0.2)	16.67 (+10.0)
	Infix	63.20 (−9.0)	64.29 (−19.9)	16.67 (+10.0)
	Suffix	55.60 (−16.6)	54.13 (−30.0)	<u>13.33</u> (+6.7)
Qwen2.5-Math-1.5B	Baseline	<u>61.40</u>	77.86	16.67
	Prefix	64.40 (+3.0)	74.30 (−3.6)	10.00 (−6.7)
	Infix	63.20 (+1.8)	73.92 (−3.9)	<u>10.00</u> (−6.7)
	Suffix	54.60 (−6.8)	58.61 (−19.3)	3.33 (−13.3)
<i>Base Models + ChatML (Format Mismatch)</i>				
Qwen2.5-Math-7B	Baseline	52.20	58.30	23.33
	Prefix	59.20 (+7.0)	56.41 (−1.9)	3.33 (−20.0)
	Infix	<u>62.00</u> (+9.8)	69.07 (+10.8)	23.33 (+0.0)
	Suffix	70.40 (+18.2)	75.97 (+17.7)	23.33 (+0.0)
Qwen2.5-Math-1.5B	Baseline	34.40	37.45	<u>13.33</u>
	Prefix	63.40 (+29.0)	67.02 (+29.6)	20.00 (+6.7)
	Infix	39.20 (+4.8)	50.42 (+13.0)	6.67 (−6.7)
	Suffix	<u>51.00</u> (+16.6)	<u>50.64</u> (+13.2)	<u>13.33</u> (+0.0)

Table 6: Number of RSP tokens (L) used for each model and benchmark.

Model	MATH-500	GSM8K	AIME24
LLaMA-3.1-8B-Instruct	15	20	15
Qwen2.5-Math-7B-Instruct	20	20	20
Qwen2.5-Math-1.5B-Instruct	20	10	20
Qwen2.5-Math-7B	10	10	20
Qwen2.5-Math-1.5B	10	10	20
Qwen2.5-Math-1.5B (ChatML)	20	20	10
Qwen2.5-Math-7B (ChatML)	20	20	20

748 A.1 RSP Injection Positions and Prompt Templates

749 Below we show the prompt structure for each injection position across all model types. Blue text
750 indicates RSP embeddings, and purple text indicates special tokens.

751 A.1.1 Prefix

752 RSP embeddings are concatenated before the prompt embeddings. No text modification is applied.

753 ChatML (Qwen Instruct / Base + ChatML).

```
[Random Embeddings (L tokens)]
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{<|im_end|>
<|im_start|>user
{question}<|im_end|>
<|im_start|>assistant
```

755 LLaMA Chat Template.

```
[Random Embeddings (L tokens)]
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date: December 2023
Today Date: 26 Jul 2024
Please reason step by step, and put your final answer within \boxed{<|eot_id|>
<|start_header_id|>user<|end_header_id|>
{question}<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

757 Raw Text (Qwen Base).

```
[Random Embeddings (L tokens)]
{question}
Please reason step by step, and put your final answer within \boxed{<|eot_id|>
```

759 A.1.2 Infix

760 A description text and special tokens are inserted into the prompt. The special token positions are
761 then replaced with RSP embeddings at the embedding level.

762 ChatML (Qwen Instruct / Base + ChatML).

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{<|im_end|>
<|im_start|>user
{question}

There are {L} special tokens that contain compressed latent reasoning information that
might be useful for your reasoning. If these tokens are useful for your case, you can
use them as reference. If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.

Here are the {L} special tokens: <special_token_1>...<special_token_L>
<|im_end|>
<|im_start|>assistant
```

764 LLaMA Chat Template.

```
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date: December 2023
Today Date: 26 Jul 2024
Please reason step by step, and put your final answer within \boxed{<|eot_id|>
<|start_header_id|>user<|end_header_id|>
```

{question}

There are $\{L\}$ special tokens that contain compressed latent reasoning information that might be useful for your reasoning. If these tokens are useful for your case, you can use them as reference. If these tokens are not useful for your case, you can ignore them and focus back to solving the problem.

Here are the $\{L\}$ special tokens:

```
<reserved_special_token_0>...  
<reserved_special_token_L>  
<|eot_id|>  
<|start_header_id|>assistant<|end_header_id|>
```

766

767 Raw Text (Qwen Base).

{question}

There are $\{L\}$ special tokens that contain compressed latent reasoning information that might be useful for your reasoning. If these tokens are useful for your case, you can use them as reference. If these tokens are not useful for your case, you can ignore them and focus back to solving the problem.

Here are the $\{L\}$ special tokens: $\langle \text{special_token_1} \rangle \dots \langle \text{special_token_L} \rangle$

Please reason step by step, and put your final answer within `\boxed{\}`.

768

769 A.1.3 Suffix

770 For chat-templated models, RSP embeddings are inserted immediately before the end-of-turn token.

771 For raw text models, RSP embeddings are concatenated at the end of the prompt.

772 ChatML (Qwen Instruct / Base + ChatML).

```
<|im_start|>system  
Please reason step by step, and put your final answer within \boxed{ }.<|im_end|>  
<|im_start|>user  
{question}[Random Embeddings ( $L$  tokens)]<|im_end|>  
<|im_start|>assistant
```

773

774 LLaMA Chat Template.

```
<|begin_of_text|>  
<|start_header_id|>system<|end_header_id|>  
Cutting Knowledge Date: December 2023  
Today Date: 26 Jul 2024  
Please reason step by step, and put your final answer within \boxed{ }.<|eot_id|>  
<|start_header_id|>user<|end_header_id|>  
{question}[Random Embeddings ( $L$  tokens)]<|eot_id|>  
<|start_header_id|>assistant<|end_header_id|>
```

775

776 Raw Text (Qwen Base).

```
{question}  
Please reason step by step, and put your final answer within \boxed{ }. [Random  
Embeddings ( $L$  tokens)]
```

777

778 A.2 Answer Extraction and Grading

779 The final answer is extracted from the model output through a fallback chain. Each step is attempted
780 in order; if extraction fails, the next step is tried.

Answer Extraction Fallback Chain

1. "final answer is \$...\$. I hope" pattern (Minerva format)
2. `\boxed{...}`: last non-empty box is used; empty `\boxed{}` are skipped
3. Text following "the answer is"
4. Text following "final answer is"
5. Last number in the output (regex fallback)

Extracted answers are normalized by removing \$, \left, \right, and unit strings. Grading is performed via symbolic equivalence checking: exact string match, numerical comparison (`math.isclose, rel_tol=1e-4`), and SymPy symbolic simplification.

A.3 Reproduced Prior Work Hyperparameters

All prior methods are reproduced on Qwen2.5-Math-1.5B-Instruct and Qwen2.5-Math-7B-Instruct, evaluated with greedy decoding (temperature = 0) and our unified answer extraction pipeline.

TTSV. Prefix length 20, trained for 20 epochs with AdamW optimizer, batch size 2, gradient accumulation 8 steps, and linear learning rate schedule. Learning rate is 1e-3 for Qwen-1.5B and 1e-5 for Qwen-7B.

LTPO. Table 7 reports the per-benchmark hyperparameters.

Table 7: LTPO hyperparameters.

Parameter	GSM8K	MATH-500	AIME24
tokens	8	8	8
steps	20	20	20
top_k	10	10	10
sigma	20	20	5
sigma_decay	0.95	0.95	0.95
lr	1e-2	5e-3	5e-2

SoftCoT. Projection module trained for 10 epochs. Thought tokens: 32 during training, 4 during evaluation. Batch size 4 for GSM8K, 1 for MATH with gradient accumulation 4. The small (assistant) model is Qwen2.5-Math-1.5B-Instruct in both configurations, while the large model is Qwen2.5-Math-1.5B-Instruct (learning rate 2e-5) or Qwen2.5-Math-7B-Instruct (learning rate 5e-6). For MATH-500 and AIME24 evaluation, we use the projection module trained on GSM8K, as it yields higher accuracy than the one trained on MATH.

A.4 Full Entropy Curves

Figure 4 shows the full normalized (0–100%) generation trajectories for entropy, top-1 probability, and varentropy, and Table 8 reports the corresponding scalar statistics including overall means, first-5% means, and the average number of generated tokens per problem. All methods converge to similar levels after the initial generation stage, confirming that the distributional changes induced by RSP are localized to the early reasoning phase.

Table 8: Per-step statistics for Qwen2.5-Math-1.5B-Instruct on MATH-500 across the full generation and the first 5% of generation steps, together with the average number of generated tokens per problem. Top-1 Prob (overall) is reported as a weighted average over the first-10%/middle-last-10% segment means with weights 0.1/0.8/0.1. Extension of Figure 1 (main text) and Figure 4.

Metric	Baseline	Prefix	Infix	Suffix	LTPO	SoftCoT	TTSV
Tokens (mean)	575.7	558.4	549.9	573.3	592.3	594.4	577.4
Entropy (first 5%)	0.1332	0.1366	0.1402	0.1941	0.1662	0.4218	0.1359
Entropy (overall)	0.0871	0.0881	0.0899	0.1049	0.0909	0.1059	0.0864
Top-1 Prob (first 5%)	0.9535	0.9523	0.9525	0.9373	0.9437	0.9156	0.9534
Top-1 Prob (overall)	0.9684	0.9681	0.9678	0.9647	0.9683	0.9651	0.9685
Varentropy (first 5%)	0.2181	0.2207	0.2463	0.4010	0.2953	2.2234	0.2174
Varentropy (overall)	0.1353	0.1375	0.1422	0.2038	0.1452	0.2404	0.1361

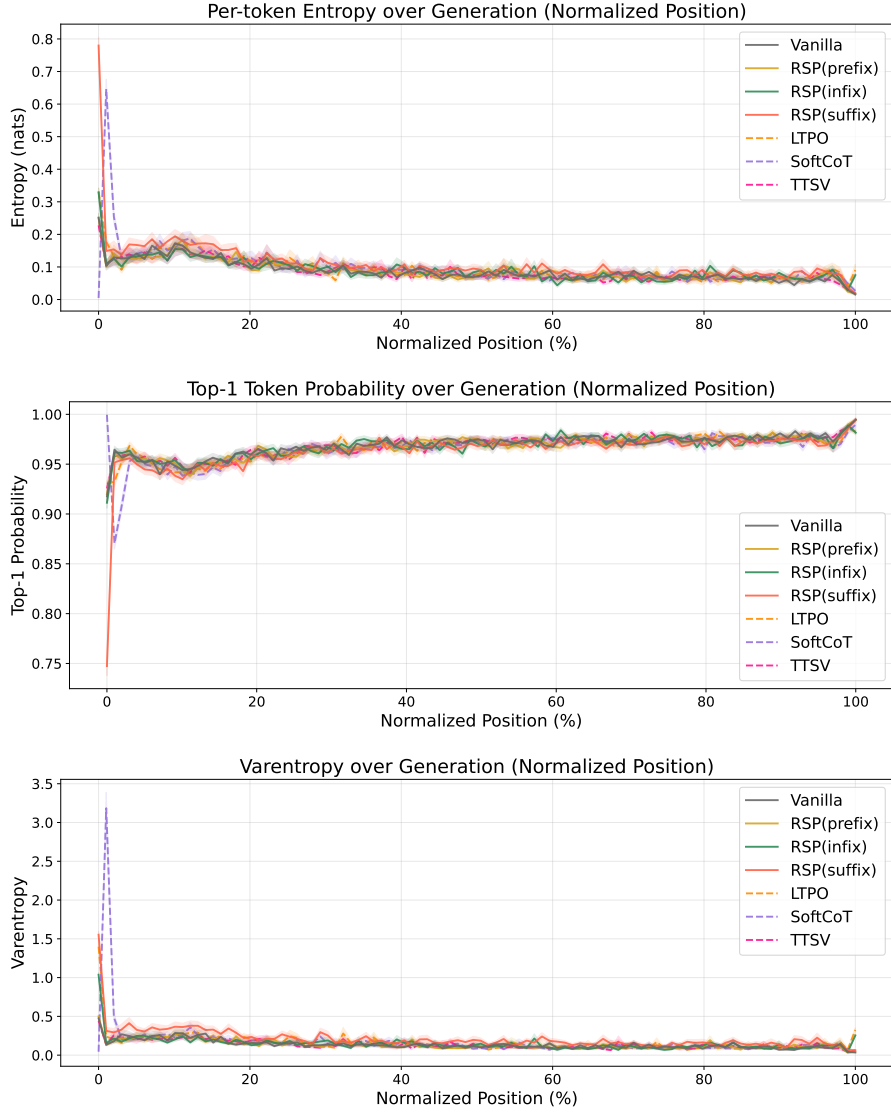


Figure 4: Mean entropy (top), top-1 probability (middle), and varentropy (bottom) over the full normalized generation trajectory (0–100%). Averaged over MATH-500, shading indicates ± 1 SEM. Solid lines: RSP variants and the baseline. Dashed lines: existing soft prompt methods. All methods converge after the initial stage.

804 B Token-Level Distribution Metrics

805 This appendix provides formal definitions for the three metrics used in the Token-level analysis of
 806 Section 3.4. Let P_{base} denote the baseline next-token distribution at $\tau = 1.0$ and P_{target} the candidate
 807 distribution being compared (e.g., the baseline at $\tau = 2.0$ or RSP at $\tau = 1.0$). All metrics are
 808 computed analytically from saved logits at the first generation step.

809 **Spearman rank correlation.** Spearman’s ρ is the Pearson correlation coefficient between the token
 810 ranks of two distributions:

$$\rho = \text{Pearson}(\text{rank}(P_{\text{target}}), \text{rank}(P_{\text{base}})). \quad (4)$$

811 Because temperature scaling $P^{(\tau)} \propto P^{1/\tau}$ is a strictly monotone transform, the rank order of tokens
 812 is preserved exactly, so $\rho = 1$ for any $\tau > 0$ as a mathematical identity. Any value $\rho < 1$ therefore
 813 indicates rank reorganization beyond what temperature scaling can produce.

814 **Mass outside top- K .** Let $\text{TopK}(P_{\text{base}})$ be the set of K tokens with the highest probability under
 815 P_{base} . The probability mass that the candidate distribution places *outside* this set is

$$\text{MassOutside}_K = 1 - \sum_{t \in \text{TopK}(P_{\text{base}})} P_{\text{target}}(t). \quad (5)$$

816 This directly measures support expansion: how much probability the candidate assigns to tokens that
 817 are not in the baseline’s preferred set. Reported values use $K = 10$.

818 **Jensen–Shannon divergence.** The symmetrized, bounded version of Kullback–Leibler divergence:

$$\text{JS}(P, Q) = \frac{1}{2} \text{KL}(P \parallel M) + \frac{1}{2} \text{KL}(Q \parallel M), \quad M = \frac{1}{2}(P + Q). \quad (6)$$

819 We use base-2 logarithms so that $\text{JS} \in [0, 1]$. Tokens absent from the stored top-100 are assigned a
 820 sentinel logit (-10^9) before softmax, which is negligible for our setting since the top-5 covers more
 821 than 99.9% of the probability mass.

C Why Random Prompts Are Semantically Neutral and Direction-Agnostic

This appendix formalizes the “why random” claim used in Section 4.1. The point is not that random prompts contain hidden task information. The point is structural: if the perturbation is meant to act as a generic exploration device rather than as a learned prompt prior, then it should be resampled across runs, centered so that it does not induce a fixed first-order drift, and direction-agnostic so that it does not under-serve whichever local branch direction matters for the current prompt.

C.1 Semantic Neutrality

Let $\bar{h} \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_d)$ and $\hat{h} := \bar{h}/\|\bar{h}\|$ whenever $\bar{h} \neq 0$. Let $u_1, \dots, u_M \in \mathbb{S}^{d-1}$ be any fixed unit vectors representing semantic anchor directions.

Lemma 3 (Semantic neutrality). *For any $\delta \in (0, 1)$,*

$$\Pr \left[\max_{r \in [M]} |\hat{h}^\top u_r| \leq \sqrt{\frac{2 \log(2M/\delta)}{d-1}} \right] \geq 1 - \delta.$$

Hence the normalized direction of an isotropic Gaussian prompt is simultaneously nearly orthogonal to every fixed finite collection of semantic anchors with high probability.

Proof. Because $\bar{h}$ is isotropic Gaussian, its normalized direction $\hat{h}$ is uniformly distributed on the sphere $\mathbb{S}^{d-1}$. For a fixed unit vector $u \in \mathbb{S}^{d-1}$, the marginal $\hat{h}^\top u$ satisfies the standard spherical concentration bound [Vershynin, 2018, Theorem 3.4.6][Boucheron et al., 2013, Section 3.2],

$$\Pr(|\hat{h}^\top u| \geq t) \leq 2 \exp\left(-\frac{(d-1)t^2}{2}\right) \quad \text{for } t \in (0, 1),$$

which follows from the spherical-cap area bound $\sigma\{\hat{h} \in \mathbb{S}^{d-1} : \hat{h}^\top u \geq t\} \leq \frac{1}{2}(1-t^2)^{(d-1)/2}$ together with $(1-t^2)^{(d-1)/2} \leq \exp(-(d-1)t^2/2)$ derived from $\log(1-x) \leq -x$ for $x \in [0, 1)$. Applying this bound to each u_r and using a union bound,

$$\Pr \left[\max_{r \in [M]} |\hat{h}^\top u_r| \geq t \right] \leq 2M \exp\left(-\frac{(d-1)t^2}{2}\right).$$

Setting

$$t = \sqrt{\frac{2 \log(2M/\delta)}{d-1}}$$

gives the claim. If the displayed value of t exceeds 1, the statement is trivial because $|\hat{h}^\top u_r| \leq 1$ always. $\square$

The contrast with a token embedding is immediate. If $e_s \neq 0$ is a fixed token embedding and $\hat{e}_s := e_s/\|e_s\|$, then choosing the anchor direction $u = \hat{e}_s$ gives

$$|\hat{e}_s^\top u| = 1.$$

A token prompt is therefore maximally aligned with at least one semantic direction, whereas an isotropic random prompt is nearly orthogonal to every fixed finite set of such directions. This is the sense in which tokens inject semantic commitment while random prompts act as semantically uncommitted perturbations.

C.2 Centered Perturbations Do Not Impose Systematic First-Order Drift

Let Δ_{ij} be the baseline logit gap between tokens i and j , and write the first-order surrogate as

$$\Delta_{ij}(\bar{h}) = \Delta_{ij} + b_{ij}^\top \bar{h}.$$

Proposition 4 (No systematic first-order drift). *If $\mathbb{E}[\bar{h}] = 0$, then*

$$\mathbb{E}[\Delta_{ij}(\bar{h})] = \Delta_{ij}$$

for every token pair (i, j) . If a deterministic prompt h_0 is reused across runs, then

$$\Delta_{ij}(h_0) - \Delta_{ij} = b_{ij}^\top h_0$$

is a fixed directional bias.

854 *Proof.* The centered case is immediate from linearity of expectation:

$$\mathbb{E}[\Delta_{ij}(\bar{h})] = \Delta_{ij} + b_{ij}^\top \mathbb{E}[\bar{h}] = \Delta_{ij}.$$

855 The deterministic case follows by substitution. $\square$

856 The proposition does not say that a centered perturbation leaves every *realization* unchanged; it says
 857 there is no systematic first-order push toward one branch across runs. This is exactly the behavior
 858 desired from a generic exploration device. A fixed token or fixed soft prompt may still be useful, but
 859 it acts as a persistent prompt prior rather than as an unbiased exploration mechanism.

860 C.3 Direction-Agnostic Coverage

861 The second requirement is directional coverage. Let D be any zero-mean perturbation distribution
 862 in $\mathbb{R}^d$ with covariance Σ_D and variance budget $\text{tr}(\Sigma_D) \leq \rho^2$. For a unit vector b , the first-order
 863 perturbation energy available along b is $\text{Var}_{h \sim D}(b^\top h) = b^\top \Sigma_D b$. This is the quantity that appears
 864 in the local linear logit surrogate used in the main text.

865 **Theorem 3** (Direction-agnostic coverage). *Define*

$$\mathcal{E}(D) := \min_{\|b\|=1} \text{Var}_{h \sim D}(b^\top h).$$

866 *Then*

$$\mathcal{E}(D) = \lambda_{\min}(\Sigma_D) \leq \frac{\rho^2}{d},$$

867 *with equality if and only if* $\Sigma_D = (\rho^2/d)\mathbf{I}_d$.

868 *Proof.* For any symmetric covariance matrix Σ_D ,

$$\min_{\|b\|=1} b^\top \Sigma_D b = \lambda_{\min}(\Sigma_D),$$

869 so $\mathcal{E}(D) = \lambda_{\min}(\Sigma_D)$. Let the eigenvalues of Σ_D be $\lambda_1, \dots, \lambda_d$. Then

$$\lambda_{\min}(\Sigma_D) \leq \frac{1}{d} \sum_{m=1}^d \lambda_m = \frac{\text{tr}(\Sigma_D)}{d} \leq \frac{\rho^2}{d}.$$

870 Equality holds if and only if all eigenvalues are equal, i.e., $\Sigma_D = (\rho^2/d)\mathbf{I}_d$. $\square$

871 This theorem is a statement about second moments, not about Gaussianity by itself. Many distributions
 872 could realize isotropic covariance. We use a Gaussian in RSP because it is the simplest centered,
 873 rotation-symmetric family that matches the scale of pretrained embeddings. The theorem clarifies the
 874 structural advantage: a fixed prompt has $\Sigma_D = 0$ and therefore no across-run exploration, whereas
 875 any anisotropic perturbation leaves at least one branch-sensitive direction less explored than the
 876 isotropic optimum.

877 **Proposition 5** (Covariance pushforward under the value projection). *Let $\bar{h}$ be a centered random*
 878 *vector with covariance $\sigma_E^2 \mathbf{I}_d$, and let $v = W\bar{h}$ for a matrix $W \in \mathbb{R}^{d_v \times d}$. Then*

$$\text{Cov}(v) = \sigma_E^2 W W^\top.$$

879 *Consequently, for every unit vector $u \in \mathbb{S}^{d_v-1}$,*

$$\sigma_E^2 \lambda_{\min}(W W^\top) \leq \text{Var}(u^\top v) = \sigma_E^2 u^\top W W^\top u \leq \sigma_E^2 \lambda_{\max}(W W^\top).$$

880 *In particular, if W is full rank then the projected perturbation has strictly positive variance in every*
 881 *output-space direction.*

882 *Proof.* Since $v = W\bar{h}$ and $\mathbb{E}[\bar{h}] = 0$,

$$\text{Cov}(v) = \mathbb{E}[v v^\top] = W \mathbb{E}[\bar{h} \bar{h}^\top] W^\top = \sigma_E^2 W W^\top.$$

883 For any unit vector u ,

$$\text{Var}(u^\top v) = u^\top \text{Cov}(v) u = \sigma_E^2 u^\top W W^\top u,$$

884 and the Rayleigh quotient bounds give the stated inequalities. $\square$

D Proofs for Attention-Mediated Annealing

D.1 Hopfield-Energy Descent and Basin Trapping

We restate the Modern Hopfield correspondence in the form used in the main text. For a key matrix $\mathbf{K} = (\mathbf{k}_1, \dots, \mathbf{k}_n)$ at a fixed query position, the Hopfield energy is

$$E_{\mathbf{K}}(\sigma) := -\beta^{-1} \log \sum_{j=1}^n \exp(\beta \mathbf{k}_j^\top \sigma) + \frac{1}{2} \|\sigma\|^2.$$

A direct calculation gives the gradient

$$\nabla E_{\mathbf{K}}(\sigma) = \sigma - \mathbf{K}^\top \text{softmax}(\beta \mathbf{K} \sigma).$$

Under (A1) $\mathbf{K} = \mathbf{V}$, the attention update with this $\mathbf{K}$ is therefore $\sigma^{\text{new}} = \sigma - \nabla E_{\mathbf{K}}(\sigma) = \mathbf{K}^\top \text{softmax}(\beta \mathbf{K} \sigma)$, which is one gradient step on $E_{\mathbf{K}}$ at unit step size [Ramsauer et al., 2021].

Lemma 4 (Descent inequality). *If $E_{\mathbf{K}}$ is L_E -smooth on a neighborhood of σ , then the unit-step gradient update $\sigma^{\text{new}} := \sigma - \nabla E_{\mathbf{K}}(\sigma)$ satisfies*

$$E_{\mathbf{K}}(\sigma^{\text{new}}) \leq E_{\mathbf{K}}(\sigma) - \left(1 - \frac{L_E}{2}\right) \|\nabla E_{\mathbf{K}}(\sigma)\|^2.$$

Proof. The standard L_E -smooth quadratic upper bound [Nesterov, 2018, Lemma 1.2.3] states $E(\sigma + v) \leq E(\sigma) + \nabla E(\sigma)^\top v + (L_E/2) \|v\|^2$ for any v in the smoothness neighborhood. Applying this to $v = -\nabla E(\sigma)$ gives $E(\sigma - \nabla E) \leq E(\sigma) - \|\nabla E\|^2 + (L_E/2) \|\nabla E\|^2 = E(\sigma) - (1 - L_E/2) \|\nabla E\|^2$. $\square$

Theorem 4 (Deterministic descent and sublevel-set invariance). *Assume (A1) $\mathbf{K} = \mathbf{V}$, (A2) the attention component is weight-tied across blocks, and (A3) $E_{\mathbf{K}}$ is L_E -smooth on a neighborhood of the unperturbed trajectory with $L_E < 2$. Then the unit-step iterate $\sigma^{(l+1)} = \sigma^{(l)} - \nabla E_{\mathbf{K}}(\sigma^{(l)})$ is strictly monotone-descending in $E_{\mathbf{K}}$ at every non-stationary point, and the sublevel set $\{\sigma : E_{\mathbf{K}}(\sigma) \leq E_{\mathbf{K}}(\sigma^{(0)})\}$ is forward-invariant.*

Proof. By Lemma 4, $E_{\mathbf{K}}(\sigma^{(l+1)}) \leq E_{\mathbf{K}}(\sigma^{(l)}) - (1 - L_E/2) \|\nabla E_{\mathbf{K}}(\sigma^{(l)})\|^2$. Under (A3), the coefficient $1 - L_E/2 > 0$, so $E_{\mathbf{K}}(\sigma^{(l+1)}) \leq E_{\mathbf{K}}(\sigma^{(l)})$ with strict inequality unless $\nabla E_{\mathbf{K}}(\sigma^{(l)}) = 0$, which proves forward invariance. $\square$

Corollary 3 (Connected-component containment for the unit-step iterate). *Let $S := \{\sigma : E_{\mathbf{K}}(\sigma) \leq E_{\mathbf{K}}(\sigma^{(0)})\}$ and let $C(\sigma^{(0)})$ denote the connected component of S containing $\sigma^{(0)}$. Define*

$$\text{sep}(C(\sigma^{(0)})) := \inf\{\|x - y\| : x \in C(\sigma^{(0)}), y \in S \setminus C(\sigma^{(0)})\},$$

with the convention $\text{sep} = +\infty$ if S is connected. Assume $\text{sep}(C(\sigma^{(0)})) > 0$. Then the unit-step iterate $\sigma^{(l+1)} = \sigma^{(l)} - \nabla E_{\mathbf{K}}(\sigma^{(l)})$ remains in $C(\sigma^{(0)})$ for all $l \geq 0$ whenever

$$\|\nabla E_{\mathbf{K}}(\sigma^{(l)})\| < \text{sep}(C(\sigma^{(0)})) \quad \text{for every } l.$$

Proof. By Theorem 4, every iterate lies in S . Suppose by induction $\sigma^{(l)} \in C(\sigma^{(0)})$. The displacement $\sigma^{(l+1)} - \sigma^{(l)} = -\nabla E_{\mathbf{K}}(\sigma^{(l)})$ has Euclidean length $\|\nabla E_{\mathbf{K}}(\sigma^{(l)})\|$. A jump to a distinct component $C' \subseteq S$ would require this length to be at least $\inf\{\|x - y\| : x \in C(\sigma^{(0)}), y \in C'\} \geq \text{sep}(C(\sigma^{(0)}))$, which the hypothesis excludes. Hence $\sigma^{(l+1)} \in C(\sigma^{(0)})$. $\square$

Positivity of sep. $\text{sep}(C) > 0$ holds whenever the components of S are spatially separated by a region of strictly higher energy than $E_{\mathbf{K}}(\sigma^{(0)})$, which we treat as a problem-dependent assumption rather than a structural property: it requires that the relevant components of the Hopfield energy do not share closure boundary points at level $E_{\mathbf{K}}(\sigma^{(0)})$. If $\text{sep} = 0$ the corollary becomes vacuous; in that case the trajectory is still in S by Theorem 4, but is not formally trapped in $C(\sigma^{(0)})$.

919 **Caveats and scope.** Theorem 4 depends on three assumptions that are individually mild but jointly
 920 idealized for real transformers. (A1) $\mathbf{K} = \mathbf{V}$ collapses the key and value projections; this is the
 921 standard Modern Hopfield assumption [Ramsauer et al., 2021] and isolates the energy-iteration aspect
 922 of attention from the additional value-projection degree of freedom. (A2) weight-tying treats the
 923 attention component as a single repeated map; without it, each block has its own energy and the
 924 trajectory does not minimize a single fixed function across blocks. (A3) is a smoothness condition on
 925 $E_{\mathbf{K}}$. For the Hopfield energy used here, $\nabla^2 E_{\mathbf{K}}(\sigma) = \mathbf{I} - \beta \mathbf{K}^\top (\text{diag}(p_\sigma) - p_\sigma p_\sigma^\top) \mathbf{K}$ where $p_\sigma =$
 926 $\text{softmax}(\beta \mathbf{K} \sigma)$, so the operator-norm bound on $\nabla^2 E_{\mathbf{K}}$ depends on the inverse-temperature β and
 927 on $\|\mathbf{K}\|$. (A3) holds whenever $\beta \|\mathbf{K}\|^2$ is sufficiently small, which is the standard regime in which the
 928 Hopfield iterate converges in a small number of steps. The theorem therefore characterizes an idealized
 929 tendency of attention-only iteration in this regime; it is not a literal property of every transformer
 930 block. In particular, sublevel-set confinement follows from Theorem 4, whereas connected-component
 931 confinement additionally requires the separation and per-step gradient assumptions of Corollary 3. Its
 932 role in the main text is to supply an idealized source of exploration deficit that an injected stochastic
 933 perturbation addresses; the empirical link is the cross-layer attention decay observed in Figure 3 and
 934 the trajectory diversity gain reported in Section 3.4.

935 D.2 Exact Attention Decomposition and Magnitude Control

936 **Proposition 6** (Attention decomposition). *Let $\alpha_{ij}^{(l)}$ be the attention weights for query position i at*
 937 *layer l , with n real tokens and k random tokens. Define*

$$w_{r,i}^{(l)} := \sum_{j=1}^k \alpha_{i,n+j}^{(l)}$$

938 *and, for $j \in [n]$,*

$$\tilde{\alpha}_{ij}^{(l)} := \frac{\alpha_{ij}^{(l)}}{1 - w_{r,i}^{(l)}}.$$

939 *Then*

$$\mathbf{x}_i^{(l+1)} = (1 - w_{r,i}^{(l)}) \tilde{\mathbf{x}}_i^{(l+1)} + \boldsymbol{\eta}_i^{(l)},$$

940 *where*

$$\tilde{\mathbf{x}}_i^{(l+1)} := \sum_{j=1}^n \tilde{\alpha}_{ij}^{(l)} \mathbf{v}_j^{(l)} \quad \text{and} \quad \boldsymbol{\eta}_i^{(l)} := \sum_{j=1}^k \alpha_{i,n+j}^{(l)} \mathbf{v}_{r_j}^{(l)}.$$

941 *Proof.* The attention output is

$$\mathbf{x}_i^{(l+1)} = \sum_{j=1}^n \alpha_{ij}^{(l)} \mathbf{v}_j^{(l)} + \sum_{j=1}^k \alpha_{i,n+j}^{(l)} \mathbf{v}_{r_j}^{(l)}.$$

942 By definition, $\sum_{j=1}^n \alpha_{ij}^{(l)} = 1 - w_{r,i}^{(l)}$, so

$$\sum_{j=1}^n \alpha_{ij}^{(l)} \mathbf{v}_j^{(l)} = (1 - w_{r,i}^{(l)}) \sum_{j=1}^n \frac{\alpha_{ij}^{(l)}}{1 - w_{r,i}^{(l)}} \mathbf{v}_j^{(l)} = (1 - w_{r,i}^{(l)}) \tilde{\mathbf{x}}_i^{(l+1)}.$$

943 Substituting this identity gives the decomposition. □

944 **Corollary 4** (Magnitude scales with random-token attention). *For every realization of the random*
 945 *values,*

$$\|\boldsymbol{\eta}_i^{(l)}\| \leq w_{r,i}^{(l)} \max_{j \in [k]} \|\mathbf{v}_{r_j}^{(l)}\|.$$

946 *Proof.* By the triangle inequality,

$$\|\boldsymbol{\eta}_i^{(l)}\| = \left\| \sum_{j=1}^k \alpha_{i,n+j}^{(l)} \mathbf{v}_{r_j}^{(l)} \right\| \leq \sum_{j=1}^k \alpha_{i,n+j}^{(l)} \|\mathbf{v}_{r_j}^{(l)}\| \leq w_{r,i}^{(l)} \max_{j \in [k]} \|\mathbf{v}_{r_j}^{(l)}\|.$$

947 □

Corollary 4 is the only quantitative fact needed in the main text. No independence assumption between attention weights and random keys is required. Once $w_{r,i}^{(l)}$ is small, the random contribution must be small as well.

D.3 Attention Decay Bound

Theorem 5 (Attention decay). *The total attention on random tokens satisfies*

$$w_{r,i}^{(l)} \leq \frac{k}{k + n \cdot \exp(\Delta_i^{(l)}/\sqrt{d})}, \quad (7)$$

where $\Delta_i^{(l)} := \min_{j \in [n]} \mathbf{q}_i^\top \mathbf{k}_j - \max_{j' \in [k]} \mathbf{q}_i^\top \mathbf{k}_{r_{j'}}$ is the real-vs-random logit gap.

Proof. Let $s_j = \mathbf{q}_i^\top \mathbf{k}_j / \sqrt{d}$ for real keys and $s_{r_{j'}} = \mathbf{q}_i^\top \mathbf{k}_{r_{j'}} / \sqrt{d}$ for random keys. By definition of $\Delta_i^{(l)}$, each real key satisfies $s_j \geq \Delta_i^{(l)}/\sqrt{d} + s_{r,\max}$ where $s_{r,\max} = \max_{j'} s_{r_{j'}}$. Therefore $\exp(s_j) \geq \exp(\Delta_i^{(l)}/\sqrt{d}) \cdot \exp(s_{r,\max})$ for every real key. Summing over n real keys gives

$$\sum_{j=1}^n \exp(s_j) \geq n \cdot \exp(\Delta_i^{(l)}/\sqrt{d}) \cdot \exp(s_{r,\max}) \geq n \cdot \exp(\Delta_i^{(l)}/\sqrt{d}) \cdot \frac{1}{k} \sum_{j'=1}^k \exp(s_{r_{j'}}).$$

Let $R := \sum_{j'=1}^k \exp(s_{r_{j'}})$. Then

$$w_{r,i}^{(l)} = \frac{R}{\sum_{j=1}^n \exp(s_j) + R} \leq \frac{R}{n \cdot \exp(\Delta_i^{(l)}/\sqrt{d}) \cdot R/k + R} = \frac{k}{k + n \cdot \exp(\Delta_i^{(l)}/\sqrt{d})}.$$

□

Corollary 5 (Sequence-wise annealing under a fixed logit gap). *For fixed k and $\Delta_i^{(l)}$, the upper bound*

$$f(n) = \frac{k}{k + n \exp(\Delta_i^{(l)}/\sqrt{d})}$$

is strictly decreasing in n . Consequently, cache growth alone decreases the maximum perturbation mass compatible with that gap.

Proof. Differentiate $f(n)$ with respect to n :

$$f'(n) = -\frac{k \exp(\Delta_i^{(l)}/\sqrt{d})}{\left(k + n \exp(\Delta_i^{(l)}/\sqrt{d})\right)^2} < 0.$$

Thus $f(n)$ is strictly decreasing. □

Corollary 6 (Vanishing random contribution under annealing). *If $n \rightarrow \infty$ along decoding while k and $\Delta_i^{(l)}$ remain fixed, then $w_{r,i}^{(l)} \rightarrow 0$. Moreover, for any fixed realization of the k random values,*

$$\|\boldsymbol{\eta}_i^{(l)}\| \leq w_{r,i}^{(l)} \max_{j \in [k]} \|\mathbf{v}_{r_j}^{(l)}\| \rightarrow 0.$$

Since the implementation samples finite-dimensional vectors and k is finite, the displayed maximum is almost surely finite.

Proof. Corollary 5 implies $w_{r,i}^{(l)} \rightarrow 0$. The bound on $\|\boldsymbol{\eta}_i^{(l)}\|$ is Corollary 4. Since the multiplier $\max_{j \in [k]} \|\mathbf{v}_{r_j}^{(l)}\|$ is finite for any fixed realization of finitely many sampled values, the product converges to zero. □

972 D.4 Feed-Forward Propagation under Local Lipschitz Control

973 A transformer block applies an attention sublayer followed by a feed-forward sublayer with residual
974 connections:

$$\mathbf{x}_i^{(l,\text{attn})} = \mathbf{x}_i^{(l)} + \text{Attn}^{(l)}(\mathbf{x}_i^{(l)}), \quad \mathbf{x}_i^{(l+1)} = \mathbf{x}_i^{(l,\text{attn})} + F^{(l)}(\mathbf{x}_i^{(l,\text{attn})}).$$

975 Let $\mathbf{x}_i^{(l,\text{attn},\text{RSP})}$ and $\mathbf{x}_i^{(l,\text{attn},\text{base})}$ denote the post-attention states under RSP and the unperturbed forward
976 pass at the same input $\mathbf{x}_i^{(l)}$.

977 **Lemma 5** (Per-block FFN propagation). *The post-attention difference satisfies*

$$\|\mathbf{x}_i^{(l,\text{attn},\text{RSP})} - \mathbf{x}_i^{(l,\text{attn},\text{base})}\| \leq \|\boldsymbol{\eta}_i^{(l)}\| + w_{r,i}^{(l)} \|\tilde{\mathbf{x}}_i^{(l+1)}\|.$$

978 If, in addition, $F^{(l)}$ is $L^{(l)}$ -Lipschitz on a bounded set containing both $\mathbf{x}_i^{(l,\text{attn},\text{RSP})}$ and $\mathbf{x}_i^{(l,\text{attn},\text{base})}$,
979 then

$$\|\mathbf{x}_i^{(l+1,\text{RSP})} - \mathbf{x}_i^{(l+1,\text{base})}\| \leq (1 + L^{(l)})(\|\boldsymbol{\eta}_i^{(l)}\| + w_{r,i}^{(l)} \|\tilde{\mathbf{x}}_i^{(l+1)}\|).$$

980 *Proof.* The unperturbed attention sublayer produces $\mathbf{x}_i^{(l,\text{attn},\text{base})} = \mathbf{x}_i^{(l)} + \tilde{\mathbf{x}}_i^{(l+1)}$, while the RSP
981 attention sublayer produces $\mathbf{x}_i^{(l,\text{attn},\text{RSP})} = \mathbf{x}_i^{(l)} + (1 - w_{r,i}^{(l)})\tilde{\mathbf{x}}_i^{(l+1)} + \boldsymbol{\eta}_i^{(l)}$ by Proposition 6. Subtracting,

$$\mathbf{x}_i^{(l,\text{attn},\text{RSP})} - \mathbf{x}_i^{(l,\text{attn},\text{base})} = \boldsymbol{\eta}_i^{(l)} - w_{r,i}^{(l)} \tilde{\mathbf{x}}_i^{(l+1)},$$

982 and the triangle inequality gives the first bound. For the second bound, write the post-FFN difference
983 as

$$[\mathbf{x}_i^{(l,\text{attn},\text{RSP})} + F^{(l)}(\mathbf{x}_i^{(l,\text{attn},\text{RSP})})] - [\mathbf{x}_i^{(l,\text{attn},\text{base})} + F^{(l)}(\mathbf{x}_i^{(l,\text{attn},\text{base})})].$$

984 By Lipschitz continuity, $\|F^{(l)}(\mathbf{x}_i^{(l,\text{attn},\text{RSP})}) - F^{(l)}(\mathbf{x}_i^{(l,\text{attn},\text{base})})\| \leq L^{(l)} \|\mathbf{x}_i^{(l,\text{attn},\text{RSP})} - \mathbf{x}_i^{(l,\text{attn},\text{base})}\|$.
985 The triangle inequality combined with the first bound gives the second. $\square$

986 **Corollary 7** (Cross-block propagation control). Define $\Delta^{(l)} := \mathbf{x}_i^{(l,\text{RSP})} - \mathbf{x}_i^{(l,\text{base})}$ with $\Delta^{(0)} = \mathbf{0}$
987 (matched inputs at layer 0), and let $L_{\text{block}}^{(l)}$ denote the local Lipschitz constant of the block map
988 $B_{\text{base}}^{(l)} : \mathbf{x} \mapsto \mathbf{x} + \text{Attn}^{(l)}(\mathbf{x}) + F^{(l)}(\mathbf{x} + \text{Attn}^{(l)}(\mathbf{x}))$ at layer l . Then

$$\|\Delta^{(l+1)}\| \leq (1 + L_{\text{block}}^{(l)}) \|\Delta^{(l)}\| + (1 + L^{(l)})(\|\boldsymbol{\eta}_i^{(l)}\| + w_{r,i}^{(l)} \|\tilde{\mathbf{x}}_i^{(l+1)}\|).$$

989 Iterating across L blocks with $\Delta^{(0)} = \mathbf{0}$ yields

$$\|\Delta^{(L+1)}\| \leq \sum_{l=0}^L \prod_{l'=l+1}^L (1 + L_{\text{block}}^{(l')}) \cdot (1 + L^{(l)})(\|\boldsymbol{\eta}_i^{(l)}\| + w_{r,i}^{(l)} \|\tilde{\mathbf{x}}_i^{(l+1)}\|).$$

990 *Proof.* Decompose the cross-block discrepancy at layer $l + 1$ into two contributions:

$$\Delta^{(l+1)} = \underbrace{B_{\text{RSP}}^{(l)}(\mathbf{x}^{(l,\text{RSP})}) - B_{\text{RSP}}^{(l)}(\mathbf{x}^{(l,\text{base})})}_{\text{(I) propagation of prior } \Delta^{(l)}} + \underbrace{B_{\text{RSP}}^{(l)}(\mathbf{x}^{(l,\text{base})}) - B_{\text{base}}^{(l)}(\mathbf{x}^{(l,\text{base})})}_{\text{(II) RSP-induced perturbation injected at block } l},$$

991 where $B_{\text{RSP}}^{(l)}$ replaces $\text{Attn}^{(l)}$ with the RSP-augmented attention. Term (I) is bounded via the block
992 Lipschitz constant: $\|(I)\| \leq L_{\text{block}}^{(l)} \|\Delta^{(l)}\|$. With residuals included one gets the factor $(1 + L_{\text{block}}^{(l)})$
993 when also propagating the residual term $\Delta^{(l)}$ itself. Term (II) is exactly the per-block injection from
994 Lemma 5, bounded by $(1 + L^{(l)})(\|\boldsymbol{\eta}_i^{(l)}\| + w_{r,i}^{(l)} \|\tilde{\mathbf{x}}_i^{(l+1)}\|)$. Combining via the triangle inequality
995 gives the recursion. The summed bound follows by unrolling the recursion: with $\Delta^{(0)} = \mathbf{0}$, induction
996 on L shows that $\|\Delta^{(L+1)}\|$ is at most the displayed sum. $\square$

997 **Boundedness of $L^{(l)}$ for standard architectures.** Modern transformer blocks combine LayerNorm,
 998 GELU or SwiGLU activations, and linear projections. LayerNorm has Lipschitz constant on the
 999 order of $\|\gamma\|/\inf\|x\|$ on every set bounded away from the origin, where γ is the LayerNorm scale
 1000 parameter [Xiong et al., 2020]. GELU and SwiGLU are smooth and therefore locally Lipschitz on
 1001 bounded activation regions, and linear projections have Lipschitz constant equal to the spectral norm
 1002 of the weight matrix. The composed self-attention sublayer has a Lipschitz constant that depends
 1003 on the spectral norms of $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V$ and the inverse-temperature β as analyzed by Kim et al.
 1004 [2021]. On the bounded activation regions traversed during a single forward pass, the per-block
 1005 Lipschitz constants in Corollary 7 are therefore finite. For a fixed model depth this yields a finite
 1006 cumulative bound on $\|\Delta^{(L+1)}\|$; stronger depth-uniform non-expansion would require additional
 1007 contraction assumptions not imposed here. This is the formal sense in which the injected perturbation
 1008 stays controlled across a finite forward pass, while the size of each new injection is set by $w_{r,i}^{(l)}$.

1009 D.5 Proofs for Output Distribution Shift

1010 The main text makes two output-level claims. Proposition 1 establishes pairwise rank randomness,
 1011 and Proposition 2 upgrades that statement to top- K entry. Both are statements about a local first-order
 1012 surrogate of the perturbed logits.

1013 D.5.1 Autoregressive Formulation

1014 In autoregressive decoding, the joint distribution over a sequence $y_{1:T}$ factorizes as

$$\Pr(y_{1:T} \mid x) = \prod_{t=1}^T p(y_t \mid x, y_{<t}),$$

1015 where each step is governed by a prefix-conditioned distribution. Any perturbation introduced by
 1016 RSP affects generation through a sequence of local changes to $p(y_t \mid x, y_{<t})$, rather than a single
 1017 global distribution.

1018 D.5.2 Local Linearization of Logits under RSP

1019 Throughout this subsection, $\bar{h} \in \mathbb{R}^d$ denotes *one* centered prompt vector $\bar{h}_l$ from Definition 1, treated
 1020 as a per-token surrogate. The actual implementation uses a length- L prompt $\mathbf{H} = (h_1, \dots, h_L)$ with
 1021 L independent draws; the propositions below identify how each individual $\bar{h}_l$ shifts the logits, and
 1022 the multi-token compounding through softmax-weighted attention is reported empirically in Figure 3
 1023 and Section 3.4. We model the perturbed logits as a function $z_i(\bar{h})$, assuming local smoothness in a
 1024 neighborhood of $\bar{h} = 0$:

$$z_i(\bar{h}) = z_i(0) + \nabla_h z_i(0)^\top \bar{h} + r_i(\bar{h}),$$

1025 where $r_i(\bar{h}) = O(\|\bar{h}\|^2)$. Defining $a_i := \nabla_h z_i(0)$, we obtain the local affine surrogate

$$z_i^{\text{lin}}(\bar{h}) := z_i + a_i^\top \bar{h}.$$

1026 This approximation is valid for sufficiently small $\|\bar{h}\|$ and is used as a first-order surrogate for the
 1027 direction of perturbation rather than as an exact global model of the transformer logits.

1028 For the two propositions below we isolate the only support property that is actually used. Let D
 1029 denote the centered prompt law. We assume

$$(S1) \quad D(U) > 0 \quad \text{for every nonempty open set } U \subseteq \mathbb{R}^d.$$

1030 The Gaussian instantiation in Section 4.1 satisfies (S1), but the proofs below require only this support
 1031 condition.

1032 D.5.3 Proof of Proposition 1 (Stochastic Ranking)

1033 *If there exists a token pair (i, j) such that $a_i \neq a_j$, and if the centered prompt law satisfies (S1), then*
 1034 $0 < \Pr_{\bar{h}}(z_i^{\text{lin}}(\bar{h}) > z_j^{\text{lin}}(\bar{h})) < 1$.

1035 *Proof.* Under the local affine surrogate,

$$z_i^{\text{lin}}(\bar{h}) - z_j^{\text{lin}}(\bar{h}) = (z_i - z_j) + (a_i - a_j)^\top \bar{h}.$$

1036 Let $b_{ij} := a_i - a_j \neq 0$. The perturbed logit difference

$$\Delta_{ij}(\bar{h}) := (z_i - z_j) + b_{ij}^\top \bar{h}$$

1037 vanishes on the affine hyperplane

$$H_0 := \{\bar{h} \in \mathbb{R}^d : (z_i - z_j) + b_{ij}^\top \bar{h} = 0\}.$$

1038 Because $b_{ij} \neq 0$, H_0 has codimension one and partitions $\mathbb{R}^d$ into the two nonempty open half-spaces

$$H_+ := \{\bar{h} : (z_i - z_j) + b_{ij}^\top \bar{h} > 0\}, \quad H_- := \{\bar{h} : (z_i - z_j) + b_{ij}^\top \bar{h} < 0\}.$$

1039 By (S1), both H_+ and H_- have positive probability, which yields

$$0 < \Pr_{\bar{h}}(\Delta_{ij}(\bar{h}) > 0) < 1.$$

1040

□

1041 D.5.4 Proof of Proposition 2 (Top-K Entry)

1042 Let $J_K(s_t)$ be the baseline top- K set at state s_t , and let $i \notin J_K(s_t)$. Define

$$\mathcal{G}_{i,K} := \{\bar{h} \in \mathbb{R}^d : \#\{j \neq i : z_j^{\text{lin}}(\bar{h}) > z_i^{\text{lin}}(\bar{h})\} \leq K - 1\}.$$

1043 If $\mathcal{G}_{i,K}$ contains a nonempty open set and the centered prompt law satisfies (S1), then

$$\Pr_{\bar{h}}(\bar{h} \in \mathcal{G}_{i,K}) > 0.$$

1044 *Proof.* Under the local affine surrogate, token i belongs to the surrogate top- K set if and only if at
 1045 most $K - 1$ tokens outrank it, i.e., if and only if $\bar{h} \in \mathcal{G}_{i,K}$. For each subset S of token indices with
 1046 $i \notin S$ and $|S| \leq K - 1$, define

$$\mathcal{O}_S := \{\bar{h} : z_j^{\text{lin}}(\bar{h}) > z_i^{\text{lin}}(\bar{h}) \text{ for all } j \in S, z_j^{\text{lin}}(\bar{h}) > z_j^{\text{lin}}(\bar{h}) \text{ for all } j \notin S \cup \{i\}\}.$$

1047 Each $\mathcal{O}_S$ is an intersection of open half-spaces and is therefore an open subset of $\mathbb{R}^d$. By construction,
 1048 every $\bar{h} \in \mathcal{O}_S$ has token i ranked $|S| + 1 \leq K$ in the affine surrogate, so $\mathcal{O}_S \subseteq \mathcal{G}_{i,K}$. Hence

$$\bigcup_{S: i \notin S, |S| \leq K-1} \mathcal{O}_S \subseteq \mathcal{G}_{i,K}.$$

1049 The reverse inclusion fails only on the measure-zero set of $\bar{h}$ that produce ties, which we do not need:
 1050 the assumption that $\mathcal{G}_{i,K}$ contains a nonempty open set is sufficient. Condition (S1) therefore implies

$$\Pr_{\bar{h}}(\bar{h} \in \mathcal{G}_{i,K}) > 0,$$

1051 which is exactly the event that token i enters the surrogate top- K set. □

1052 D.5.5 Proof of Corollary 2 (Sample Complexity)

1053 Let $\mu_i = \Pr_{\bar{h}}(\bar{h} \in \mathcal{G}_{i,K}) > 0$. Within N independent RSP samples $\bar{h}_1, \dots, \bar{h}_N$, the probability of at
 1054 least one admission is at least $1 - (1 - \mu_i)^N \geq 1 - \exp(-N\mu_i)$.

1055 *Proof.* The samples $\{\bar{h}_n\}_{n=1}^N$ are independent, so

$$\Pr(\text{no admission within } N \text{ samples}) = \prod_{n=1}^N \Pr(\bar{h}_n \notin \mathcal{G}_{i,K}) = (1 - \mu_i)^N.$$

1056 The complement gives the first inequality. The second inequality is standard: $1 - \mu_i \leq \exp(-\mu_i)$,
 1057 hence $(1 - \mu_i)^N \leq \exp(-N\mu_i)$. □

1058 D.5.6 Trajectory-Level Implications

1059 Since autoregressive generation composes conditional distributions across steps, perturbations at
 1060 early steps can lead to distinct trajectories. Let $\mathcal{R}_{k,K}(x)$ denote the set of trajectories containing
 1061 at least one token outside the baseline top- K set. If RSP increases the accessibility of such tokens
 1062 at early steps, then the probability of sampling trajectories in $\mathcal{R}_{k,K}(x)$ increases under repeated
 1063 sampling. The key point is temporal compounding: one early top- K entry changes the next decoding
 1064 state, which changes the next candidate set, and so on. This is the formal bridge from local rank
 1065 perturbations to global trajectory diversity.

E Attention Mass Analysis

This appendix formalizes the per-token attention mass reported in Figure 3 and the exclusion criteria applied to the reasoning span and the layer axis. For each of the 500 MATH-500 problems, a fresh RSP $\mathbf{H} \in \mathbb{R}^{L \times d}$ with $L = 10$ is sampled independently, so the reported heatmaps average over both problem variability and RSP-vector variability.

E.1 Per-Token Attention Mass

For each problem, we measure attention on the concatenated sequence [prompt; $\mathbf{H}$; generation], where the generation is the trajectory produced by the same model under matching RSP injection. Let $\alpha_{i,j}^{(\ell,h)}$ denote the post-softmax attention weight at layer ℓ , head h , from query position i to key position j , satisfying $\sum_j \alpha_{i,j}^{(\ell,h)} = 1$ under the causal mask. Let Q and R denote the index sets of question and RSP tokens with sizes $|Q|$ and $|R| = L$, and let H be the number of heads. The per-token attention mass that query i directs to each group at layer ℓ is

$$\text{PTM}_Q[\ell, i] = \frac{1}{|Q|H} \sum_{h=1}^H \sum_{j \in Q} \alpha_{i,j}^{(\ell,h)}, \quad \text{PTM}_R[\ell, i] = \frac{1}{|R|H} \sum_{h=1}^H \sum_{j \in R} \alpha_{i,j}^{(\ell,h)}. \quad (8)$$

Normalization by $|Q|$ and $|R|$ controls for group size, so both quantities express how much attention a single question (resp. RSP) token receives on average under the same softmax denominator. Question tokens thereby serve as a size-matched baseline that absorbs sequence-length effects common to both groups.

E.2 Heatmap Aggregation

We partition the layer indices and the reasoning position indices each into five equal-size bins $\{B_L^{(b)}\}_{b=1}^5$ and $\{B_P^{(b)}\}_{b=1}^5$. For sample s and target group $\bullet \in \{Q, R\}$, the value in cell (b_L, b_P) is

$$M_{\bullet}^{(s)}[b_L, b_P] = \frac{1}{|B_L^{(b_L)}| \cdot |B_P^{(b_P)}|} \sum_{\ell \in B_L^{(b_L)}} \sum_{i \in B_P^{(b_P)}} \text{PTM}_{\bullet}[\ell, i]. \quad (9)$$

Figure 3 reports the sample average of Eq. (9) over the 500 valid samples.

E.3 Excluding the `\boxed{...}` Span

The reasoning position range terminates strictly before the final `\boxed{...}` span. Chain-of-thought outputs decompose into a *text* (content) component and a *pattern* (template) component that contribute independently to downstream performance [Madaan and Yazdanbakhsh, 2022]. The `\boxed{...}` wrapper is a canonical pattern: a short, stereotyped span whose role is to complete the answer format rather than to extend reasoning. Including it would therefore inject a template-driven attention signature unrelated to reasoning dynamics.

E.4 Excluding the Final Two Layers

The layer axis further excludes the last two transformer layers. Two well-documented properties of late layers, none specific to our setup, motivate this choice.

Output-projection specialization. Late-layer hidden states lie in an approximately linear relationship with vocabulary logits and therefore function as a progressive linear readout rather than as representation-transforming blocks [Belrose et al., 2023]. Consistently, late-layer feed-forward modules have been characterized as output-distribution shaping mechanisms [Geva et al., 2021]. Attention in these layers therefore reflects output formation rather than reasoning computation.

Register-token hypothesis. Uninformative patch tokens in Vision Transformers have been shown to be repurposed as *registers*: storage slots that absorb attention in deeper layers [Darcet et al., 2024]. Because RSPs share a similar profile as continuous tokens without pretrained semantic content, a

1104 similar repurposing in late transformer layers cannot be ruled out. Excluding the final two layers
 1105 removes this potential register-induced confound from the reasoning-phase signal.

1106 The two lines of evidence above identify known late-layer phenomena — linear readout and potential
 1107 register repurposing — whose attention signatures are orthogonal to reasoning-phase dynamics.
 1108 Removing the final two layers preempts their contamination of the reasoning-phase signal, yielding a
 1109 principled, literature-grounded exclusion rather than a model-specific tuning decision.

1110 E.5 Additional Suffix Heatmaps

1111 Figure 5 reports the same per-token RSP attention analysis under suffix injection for Qwen2.5-Math-
 1112 1.5B-Instruct and LLaMA-3.1-8B-Instruct. For the late-layer reasons discussed above, the pattern
 1113 does not generalize uniformly across every cell; nevertheless, the overall trend is consistent across
 1114 both models: question-token attention varies broadly across layers, whereas RSP attention shows the
 1115 layer-wise and sequence-wise decay predicted by Eq. (3).

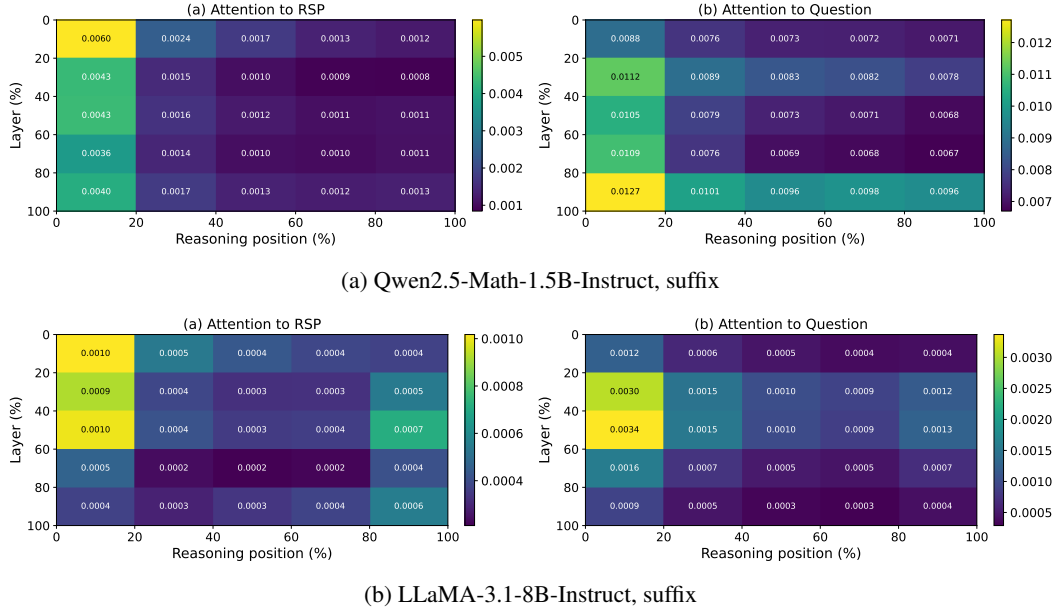


Figure 5: Per-token RSP attention mass under suffix injection for the remaining two models (500 MATH-500 samples each). Axes and preprocessing match Figure 3: x-axis is reasoning position binned into five quantiles, y-axis is layer depth binned into five quantiles (top = shallow), color encodes the 500-sample mean per-token attention assigned to RSP tokens, and tokens inside `\boxed{}` spans and the last two layers are excluded.

1116 **F GRPO Formulation**

1117 For a given prompt q , the behavior policy $\pi_{\theta_{\text{old}}}$ samples a group of G responses $\{o_i\}_{i=1}^G$ with
 1118 corresponding rewards $\{R_i\}_{i=1}^G$. The advantage of each response is computed by normalizing the
 1119 group-level rewards:

$$\hat{A}_{i,t} = \frac{R_i - \text{mean}(\{R_i\}_{i=1}^G)}{\text{std}(\{R_i\}_{i=1}^G)}. \quad (10)$$

1120 The policy is updated via a clipped objective with a KL penalty:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \left(\min \left(r_{i,t} \hat{A}_{i,t}, \text{clip}(r_{i,t}, 1-\varepsilon, 1+\varepsilon) \hat{A}_{i,t} \right) - \beta D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}) \right) \right], \quad (11)$$

1121 where $r_{i,t} = \pi_{\theta}(o_{i,t} \mid q, o_{i,<t}) / \pi_{\theta_{\text{old}}}(o_{i,t} \mid q, o_{i,<t})$ is the importance sampling ratio, ε is the clipping
 1122 range, and β controls the strength of the KL penalty against the reference policy π_{ref} .

1123 G Broader Impacts

1124 This work provides empirical and theoretical analysis of how random embedding injection affects
1125 the reasoning behavior of large language models. The contributions are primarily foundational,
1126 with potential downstream implications for reinforcement learning-based post-training of reasoning
1127 models such as GRPO.

1128 **Positive impacts.** RSP-induced trajectory diversity could improve the sample efficiency of RL-
1129 based LLM post-training by providing richer learning signals within group-relative advantage estima-
1130 tion. This has the potential to reduce compute costs associated with alignment and reasoning-oriented
1131 fine-tuning, making such methods more accessible to research groups with limited resources.

1132 **Potential concerns.** Methods that expand the effective output support of LLMs may increase the
1133 reachability of low-probability tokens, which includes both desirable alternative reasoning paths
1134 and potentially unsafe or off-distribution outputs. Practitioners applying RSP in production systems
1135 should combine it with existing safety filtering and alignment mechanisms rather than treating it as a
1136 standalone diversity source.

1137 **Limitations on impact scope.** Since this work focuses on analysis rather than deploying a new
1138 model or dataset, the direct societal impact is limited. The broader implications described above are
1139 contingent on future work integrating RSP into applied training pipelines.

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1337 Justification: **[TODO]**

1338 Guidelines:

- 1339 • The answer [N/A] means that the paper does not include experiments.
- 1340 • The paper should indicate the type of compute workers CPU or GPU, internal cluster,
1341 or cloud provider, including relevant memory and storage.
- 1342 • The paper should provide the amount of compute required for each of the individual
1343 experimental runs as well as estimate the total compute.

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1346 didn't make it into the paper).

1347 **9. Code of ethics**

1348 Question: Does the research conducted in the paper conform, in every respect, with the
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1350 Answer: **[TODO]**

1351 Justification: **[TODO]**

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1362 Answer: **[TODO]**

1363 Justification: **[TODO]**

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